FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Caesarean Section

Tutorial | Week 2 · Labour, delivery & the puerperium | Obstetrics | Dr Ndiwalana

Caesarean section (CS) is delivery of the baby through incisions in the abdominal wall and uterus. It saves lives when vaginal birth is impossible or dangerous, but it carries real surgical risk and a scar that shapes every future pregnancy. So the skill is choosing the right woman, at the right urgency, with safe preparation, technique, aftercare and a clear plan for the next birth.

Must know before the ward round

1. Urgency categories (NICE/Lucas). Category 1 is an immediate threat to the life of the woman or fetus, with birth within 30 minutes of the decision. Category 2 is compromise that is not immediately life-threatening, with birth within 75 minutes. Category 3 needs early birth but there is no compromise. Category 4 is elective.
2. The Robson Ten-Group Classification (WHO standard since 2015) sorts every woman who gives birth into 1 of 10 groups using parity, previous CS, onset of labour, gestation, fetal number and presentation. Use it to audit CS rates, not to decide on one woman.
3. Give prophylactic antibiotics BEFORE skin incision (within 60 minutes), not after cord clamping. WHO/ACOG use cefazolin 2 g IV or ampicillin 2 g IV. Most Ugandan hospitals use ceftriaxone IV (1–2 g) with or without metronidazole 500 mg IV. It is a single dose.
4. Always listen to the fetal heart in theatre just before incision, and in the second stage re-examine vaginally. The situation may have changed (IUFD, or the head is now low enough for vaginal delivery).
5. The lower-segment transverse incision is the standard. A classical (vertical upper-segment) incision carries a 4–9% rupture risk in later pregnancies, so every future birth must be by planned CS before labour.
6. VBAC after one lower-segment CS succeeds in about 72–75% of women (85–90% if she has had a vaginal birth before). The uterine rupture risk is about 0.5% (1 in 200), and induction or augmentation increases it 2–3 fold. Misoprostol is contraindicated for induction when there is a uterine scar.
7. The most consistent early sign of scar rupture is an abnormal fetal heart (bradycardia or decelerations). Maternal tachycardia, new constant pain, vaginal bleeding, haematuria, loss of station and cessation of contractions follow.
8. ERAS means early feeding, early mobilisation, early catheter removal, multimodal analgesia and thromboprophylaxis. It shortens stay without increasing complications.
9. Pre-op sign-off before every CS. Indication and category written with the decision time, consent, anaesthetic review, Hb and cross-match (2 units; 4 or more for praevia, abruption, rupture or accreta), two 16–18 G cannulae, Foley catheter, antibiotic within 60 minutes before incision, and the WHO checklist Time out.
10. A spinal needs platelets above about 75–80 × 10^9/L and no coagulopathy. Use the bedside clotting test (no clot by 7 minutes means coagulopathy) where lab clotting is unavailable.
11. After CS, remove the catheter once she is mobile and at least 12 h after the last spinal dose, but keep it 7–14 days after a bladder repair or obstructed labour with haematuria. Give enoxaparin 40 mg SC daily, first dose at least 4 h after the spinal.
12. Post-CS fever: work through womb (endometritis, day 2–5), wound (SSI, day 4–7), water (UTI), wind (chest), walking (VTE), breasts, cannula sites and malaria. Culture before antibiotics, and drain pus rather than just escalating antibiotics.

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1. Purpose and Indications

A **caesarean section (CS)** is the delivery of a baby (after fetal viability) through an incision in the abdominal wall (**laparotomy**) and the uterus (**hysterotomy**). In Uganda it is the operation that ends **obstructed labour**, rescues babies in **fetal distress**, and delivers women with **placenta praevia, cord prolapse and ruptured uterus**. It is one of the two **CEmONC signal functions** that distinguish an HC IV or hospital from an HC III. The other is blood transfusion.

But a CS is **major abdominal surgery**. It carries risks of haemorrhage, sepsis, VTE and bladder injury, and anaesthetic death. It also leaves a **scar**, and the scar raises the risk of **uterine rupture and placenta accreta** in every future pregnancy. That matters enormously in a country where women often want 5 or more children and may labour far from a theatre.

The one thing never to forget: a CS decision has two parts: *is it indicated?* and *how urgent is it?* Then everything, including consent, antibiotics before incision, the fetal heart check in theatre, a safe technique and good aftercare, follows a checklist. And **every woman leaves with a plan for her next birth.**

Definitions

Term	Meaning
Elective (planned) CS	Scheduled before labour, ideally at **≥39 weeks** to reduce neonatal respiratory problems (transient tachypnoea)
Emergency CS	Decided on because of a new problem in labour or pregnancy. The urgency is graded 1–3
Lower-segment CS (LSCS)	Transverse incision in the thin, less vascular lower uterine segment. It is the standard operation
Classical CS	Vertical incision in the **upper (contractile) segment**
De Lee / low vertical	Vertical incision confined to the lower segment. It is rare
Scar dehiscence	Separation of the old scar with the **serosa (visceral peritoneum) intact**. It is often silent and found at repeat CS as a "window"
Uterine rupture	**Full-thickness** tear with communication into the peritoneal cavity. The baby or placenta may be extruded
TOLAC / "trial of scar"	Trial of labour after caesarean. "Trial of scar" is the common Ugandan ward term
VBAC	Vaginal birth after caesarean: a *successful* TOLAC
ERCS	Elective repeat caesarean section

Indications: absolute vs relative

Absolute (vaginal birth impossible or clearly dangerous)	Relative (CS is safer in *this* situation)
Major placenta praevia (placenta covering or near the os)	Fetal distress in the first stage (non-reassuring FHR)
True CPD / contracted pelvis; **obstructed labour with a live fetus**	Breech (especially primigravida, footling, EFW >3.5–3.8 kg or <2 kg), unless the criteria for vaginal breech are met
Transverse or oblique lie in labour	Previous CS ×1 with an unfavourable picture; failed TOLAC
Previous classical, T- or J-incision, or previous uterine rupture	**Two or more previous CS** (Ugandan standard practice)
Cord prolapse with a live fetus, not fully dilated	Failed induction; arrest of dilatation or descent despite adequate contractions
Pelvic mass obstructing the birth canal (lower-segment fibroid, ovarian tumour)	Severe PE or eclampsia with an unfavourable cervix and delivery needed urgently
Invasive cervical cancer	APH (abruption with a live fetus; minor praevia bleeding)
Previous successful **VVF/RVF repair** or major pelvic floor reconstruction	Twins with first twin non-cephalic; monoamniotic twins
Brow presentation, face mento-posterior at term in labour	Macrosomia (EFW >4.5 kg in diabetes); maternal medical disease (e.g. selected cardiac); HIV with high viral load near term (individualise)

Urgency classification (NICE NG192; Lucas)

Category	Definition	Examples	Decision-to-delivery target
1	Immediate threat to the life of the woman or fetus	Cord prolapse with bradycardia, suspected uterine rupture, major abruption with a live fetus, prolonged fetal bradycardia	**As soon as possible, within 30 min**
2	Maternal or fetal compromise that is **not** immediately life-threatening	Obstructed labour, non-reassuring FHR that is not pre-terminal, APH that is controlled	**Within 75 min**
3	No compromise but needs **early** delivery	Failed induction with a well baby, woman booked for elective CS who arrives in early labour	Within hours
4	Timed to suit the woman and the team	Elective CS for 2 previous scars or breech	Scheduled

> **EXAM FAVOURITE**
>
> The urgency category describes **the patient's clinical state**, not how busy the theatre is. When you call theatre, say the category out loud ("Category 1 CS for cord prolapse with bradycardia"). Everyone then knows the clock has started.

Robson Ten-Group Classification (WHO 2015/2017). It uses 6 variables: **parity, previous CS, onset of labour, gestational age, number of fetuses, and presentation/lie.**

Group	Women included
1	Nulliparous, single cephalic, ≥37 wk, **spontaneous labour**
2	Nulliparous, single cephalic, ≥37 wk, **induced** (2a) or **CS before labour** (2b)
3	Multiparous **without** previous CS, single cephalic, ≥37 wk, spontaneous labour
4	Multiparous without previous CS, single cephalic, ≥37 wk, induced (4a) or CS before labour (4b)

Group	Women included
5	**All multiparous with ≥1 previous CS**, single cephalic, ≥37 wk
6	All **nulliparous breeches**
7	All **multiparous breeches** (including previous CS)
8	All **multiple pregnancies** (including previous CS)
9	All **transverse or oblique lies** (including previous CS)
10	All single cephalic **<37 weeks** (including previous CS)

CLINICAL PEARL

Why Robson? A crude CS rate ("our rate is 30%") tells you nothing about *why*. Robson groups let a hospital see which groups drive its rate. Usually **Group 5 (previous CS)** is the biggest, followed by Groups 1 and 2. That identifies where to intervene: safer TOLAC, better labour management in first-time mothers, fewer unnecessary inductions. The groups are **mutually exclusive and totally inclusive**: every woman fits exactly one.

Burden and Uganda context

- The WHO has long said that population CS rates **above about 10–15%** are not associated with further falls in maternal or newborn mortality. Rates **below about 5%** suggest women who need CS are not getting it.
- Uganda's national CS rate is roughly **7%** (UDHS 2022). It is rising, but it is **very unequal**: urban, wealthier and educated women have much higher rates, private hospitals in Kampala exceed 30%, and many rural districts remain below 5%. Referral hospitals such as Mulago/Kawempe, Mengo and Mbarara have rates of **30–40%**, because they receive the complicated referrals.
- HC IVs are meant to provide CEmONC (CS and transfusion). Many are limited by absent anaesthetic staff, no blood, power or water problems, or no theatre running at night.
- Leading indications in Ugandan hospitals: **previous CS, obstructed labour/CPD, fetal distress, malpresentation, APH, and hypertensive disorders.**
- **Maternal mortality after CS** in sub-Saharan Africa is many times higher than in high-income countries (about 50 times, in a 2019 Lancet Global Health meta-analysis). Most deaths come from **haemorrhage, sepsis and anaesthetic complications**, often because the operation was done late on an exhausted, infected, dehydrated woman after a long referral.
- **Uterine rupture** causes a significant share of maternal deaths in Uganda. Previous CS, obstructed labour, grand multiparity, unsupervised use of oxytocin or local herbs (*"emumbwa"*), and living far from a facility are recurring risk factors in Mulago and Mbarara studies.
- **Surgical site infection** after CS in Uganda is common (10–15% or more in several studies). Isolates at Mulago are often **ceftriaxone-resistant *Klebsiella*, *E. coli* and MRSA**. This is a strong argument for correct prophylaxis timing and asepsis rather than prolonged post-op antibiotics.

UGANDA CONTEXT

Uganda's reality creates two opposite dangers. **Too few CS** means rural women die of obstructed labour and rupture. **Unnecessary CS** means a young woman with a scar may later labour at home or at an HC III, far from a theatre, and rupture. So every CS decision in Uganda should consider: *where will this woman deliver next time?*

2. Contraindications and Precautions

Is the baby alive? A CS for a **dead baby** exposes the mother to all the risk for none of the benefit. A **vaginal birth** is usually safer for her. Exceptions are praevia, transverse lie with obstruction, rupture, or failed destructive options. Always re-check the FHR (Pinard or Doppler) and, where available, confirm IUFD by ultrasound before committing.

Stabilise first when the mother is the patient: an eclamptic woman gets **$MgSO_4$ and BP control before** she goes to theatre; a shocked woman gets resuscitation *while* theatre is prepared. A dead or dying mother cannot be helped by a fast CS on an unstable circulation (except perimortem CS during arrest).

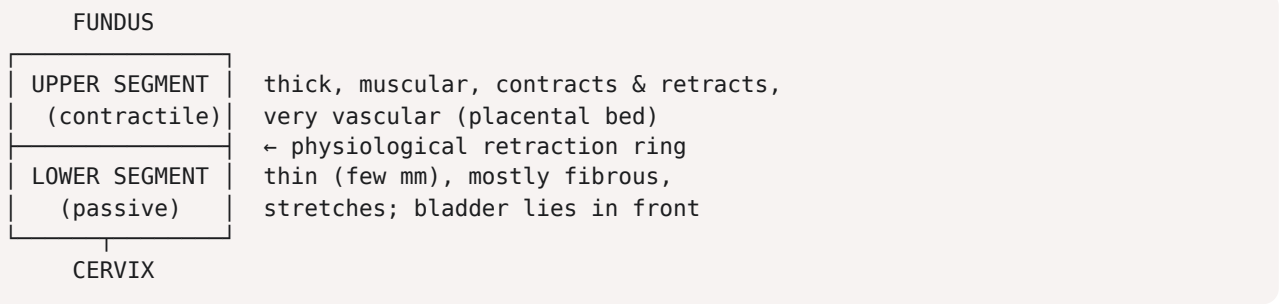
Contraindications to spinal anaesthesia: patient refusal, **coagulopathy** (DIC in abruption, HELLP with platelets <75–80 × 10^9/L), **uncorrected hypovolaemia/shock** (ruptured uterus, massive APH), local sepsis or deformity at the site, raised intracranial pressure, and some cardiac lesions (fixed cardiac output). These women need **GA** with an experienced anaesthetist.

Precautions

Situation	Precaution
Suspected placenta praevia	**No digital VE outside theatre** (double set-up only)
Previous classical, T- or J-incision	Absolute contraindication to labour; elective CS at 36–37 weeks
Prophylactic LMWH	A spinal is usually delayed until at least **12 h after a prophylactic LMWH dose**
$MgSO_4$ running	Potentiates muscle relaxants: tell the anaesthetist
Last meal	Aspiration risk under GA; elective CS fasting about **6 h for solids and 2 h for clear fluids** (unit practice varies)
Allergies (penicillin, latex, iodine, chlorhexidine)	Antibiotic and skin-prep choice
Severe PE with renal impairment, asthma, PPH with oliguria, peptic ulcer	Avoid NSAIDs post-op
Views on blood transfusion (e.g. a Jehovah's Witness)	Must be known and documented before, not during, a haemorrhage
HIV-positive or hepatitis B	Double gloving and extra sharps care; eMTCT plan for the baby
Fever before surgery	Cultures, then a **therapeutic** antibiotic course (not a single prophylactic dose)
Oxytocin running with an abnormal FHR or suspected rupture	Stop it

3. Relevant Anatomy and Normal Values

Why the lower segment? In late pregnancy and labour the uterus has two functional parts:



- The **lower segment** forms from the isthmus from about 28 weeks and thins further in labour. It is **thin, less vascular and "quiet"** (it hardly contracts), so an incision there bleeds less, heals as a stronger scar, and is less likely to rupture in later labour (about **0.5%**, versus **4–9%** for a classical scar).
- The lower segment is **covered by loose peritoneum (the uterovesical fold)**, which can be peeled down with the bladder. This keeps the bladder out of the way and keeps the scar extraperitoneal.
- In **obstructed labour** the lower segment becomes **very thin and stretched**, and the bladder is **drawn up** onto the abdomen. This is why bladder injury and lateral extensions into the uterine vessels are much more common in Ugandan emergency CS for obstructed labour. A pathological **Bandl's ring** is the visible retraction ring of impending rupture.
- A **very preterm uterus (<28–30 weeks)** has no well-formed lower segment. This is one reason a classical or low vertical incision is sometimes needed.
- The uterus is usually **dextrorotated** (rotated to the right): check that the round ligaments are equidistant before incising, so the incision is central and does not run into the left uterine vessels.
- **Abdominal wall layers**: skin, fat, rectus sheath, rectus muscles (separated in the midline), parietal peritoneum, then the uterovesical fold and the uterus.

Why antibiotics *before* incision? The antibiotic must be in the tissues when bacteria arrive, which is at incision. Giving it after cord clamping (the old practice, to "protect the baby") was shown to cause more maternal endometritis and wound infection without benefiting the baby.

Why a left lateral tilt? In the supine position the gravid uterus compresses the **inferior vena cava** (less venous return, so hypotension) and the aorta. After spinal anaesthesia (sympathetic block, so vasodilatation) this can cause profound hypotension and fetal hypoxia. Tilting 15° to the left moves the uterus off the vessels.

Normal values

Item	Normal / threshold
Fetal heart rate	**110–160 bpm**; listen for a full minute after a contraction
Shock index (HR ÷ SBP)	**≥0.9 abnormal**; **≥1.4 needs urgent intervention**
Maternal pulse post-op	**<100**
Urine output	**≥30 mL/h**
Hb	**≥11 g/dL** normal in pregnancy; **<7 g/dL severe**
WBC in labour	Up to about **25 × 10^9/L** can be normal
Platelets for a spinal	Above about **75–80 × 10^9/L**
Fibrinogen	**<2 g/L** = severe coagulopathy
Bedside clotting test	Clot by **7 minutes** (no clot = coagulopathy)
Random blood sugar	**>11 mmol/L** is high
Creatinine	**>90 µmol/L** is raised in pregnancy
Fundus day 1 post-op	Firm, at or below the umbilicus, mildly tender
Fasting before elective CS	About **6 h solids, 2 h clear fluids**
Moulding	Graded **+1 to +3**; head palpable in **fifths (5/5 to 0/5)**

4. Equipment and Preparation

Equipment: a CS set (scalpels, Mayo scissors, Kocher's forceps, artery forceps, **Doyen retractor**, **Green-Armytage** clamps, Babcock forceps, needle holders, swabs counted), **#1 absorbable suture** (polyglactin/Vicryl or chromic catgut) for the uterus and sheath, 3-0 Vicryl/Monocryl or nylon or staples for skin, suction, diathermy, chlorhexidine-alcohol or povidone-iodine, Foley catheter (14–16 Fr) and bag, two 16–18 G cannulae, crystalloid, cross-matched blood, spinal set and hyperbaric bupivacaine, oxytocin, a **resuscitaire with bag-valve-mask** and warm towels for the baby.

Pre-operative assessment: history

When you clerk a woman who may need a CS, or a woman with a previous scar in labour, every question has a purpose:

Question	Why it matters
Gravidity, parity, gestational age (LNMP and early scan)	Term or preterm? Robson group? Preterm changes the incision and the neonatal plan
The indication in one line; absolute or relative?	A relative indication (e.g. "slow progress") should make you ask whether augmentation or a ventouse is possible first
Number of previous CS, their indications, and where done	A **recurrent** indication (CPD, contracted pelvis) predicts a failed TOLAC. A non-recurrent one (breech, fetal distress) predicts success. **Two or more scars** means elective CS in most Ugandan units
Type of uterine incision (ask for the operation note or discharge form)	A **classical or T-shaped incision is an absolute contraindication to labour** and changes the incision, the urgency and the next-pregnancy plan. The skin scar does NOT tell you the uterine incision
Post-op complications after the last CS (fever, wound sepsis, long admission, re-operation)	Post-op sepsis may have weakened the scar and raises the rupture risk
Inter-delivery interval	Less than 18 months raises the rupture risk
Any vaginal birth before or after the CS	The single best predictor of VBAC success
Onset, duration and strength of labour; time membranes ruptured; colour of liquor	Prolonged labour or ROM increases infection risk and the need for therapeutic antibiotics. Meconium suggests fetal compromise
Pushing at home or at a lower facility; traditional herbs (e.g. *emumbwa*); oxytocin given elsewhere	These are major rupture risks; herbal oxytocics cause hyperstimulation
Pain character: constant between contractions? shoulder-tip pain?	Constant pain suggests abruption or rupture. Shoulder-tip pain means haemoperitoneum irritating the diaphragm
Bleeding (APH), **fetal movements**	Praevia or abruption. Absent movements may mean IUFD, which changes the decision (a vaginal route is usually preferred for a dead baby)
Medical history: hypertension, diabetes, sickle cell, cardiac disease, asthma, bleeding disorders, **HIV status and viral load; hepatitis B**	Anaesthetic and drug choices (e.g. avoid NSAIDs in severe PE with renal impairment), eMTCT plan, VTE risk, sharps care
Current drugs: $MgSO_4$, antihypertensives, anticoagulants, ARVs	LMWH timing for spinal; $MgSO_4$ potentiates muscle relaxants
Previous anaesthetic problems	Anaesthetic plan
Time of last meal and last drink	Aspiration risk if general anaesthesia is needed
Allergies (penicillin, latex, iodine, chlorhexidine)	Antibiotic and skin-prep choice
Views on blood transfusion	Must be documented before theatre

Question	Why it matters
Future fertility wishes; contraception	Offer **tubal ligation** or **PPIUD** with proper, separate consent before theatre. Plan the next birth

Pre-operative assessment: examination

Follow the order of Hutchison's Clinical Methods (24th edition). In a Category 1 situation, the focused ABC, fetal heart and abdominal check (section 7) come first; do not delay delivery for a full examination.

1. **Introduction, identity, explanation and consent**; chaperone for the vaginal examination; **hand hygiene**.
2. **Position and exposure**: semi-recumbent with a **left lateral tilt** (avoid supine hypotension); expose the abdomen from xiphisternum to symphysis; keep her covered otherwise.
3. **General inspection and vital signs**: pallor (Hb for transfusion planning), **dehydration** (dry tongue, sunken eyes, **ketones** after a long labour), fever (chorioamnionitis), oedema, **weight/BMI** (antibiotic dose, choice of incision, VTE risk). **Pulse and BP** (tachycardia is the earliest sign of haemorrhage or sepsis; hypertension points to PE), RR, temperature, SpO_2, shock index.
4. **Hands, face and neck; airway and back** (for the anaesthetist, but you should notice): palmar and conjunctival pallor; obesity, short neck, limited mouth opening; skin infection or deformity over the lumbar spine (contraindicates a spinal at that site).
5. **Chest and heart**: wheeze, crackles, murmurs.
6. **Obstetric abdominal examination**:
   - **Inspection**: scars (Pfannenstiel, midline) and their healing; distension; a **Bandl's ring** (a groove running obliquely across the abdomen).
   - **Palpation**: **tenderness over the scar** (a late and unreliable sign of dehiscence); **fundal height**; **lie**; **presentation**; **engagement in fifths palpable** (5/5 to 0/5): a head that is **3/5 or more palpable** after a long labour, with moulding, means CPD; a tender, tense lower segment with Bandl's ring means **impending rupture**; contractions (frequency and strength); easily palpable fetal parts (rupture).
   - **Auscultation of the fetal heart** for a full minute after a contraction.
7. **Bladder**: distended? **Haematuria** on catheterisation suggests obstructed labour or rupture involving the bladder.
8. **Vaginal examination** (not if placenta praevia is suspected, until in theatre with a double set-up): cervical dilatation, **station, position, caput and moulding (graded +1 to +3)**, liquor colour, cord presentation or prolapse, oedematous cervix or vulva (obstruction), and the pelvis (sacral promontory, ischial spines, subpubic angle). If she is **fully dilated**, repeat the VE in theatre: the head may now be low enough for a ventouse.
9. **Complete**: thank her, cover her, hand hygiene, **summarise** with the indication and urgency category, and document positives and key negatives.

Pre-operative investigations (what to send)

Hb and group & cross-match for all (2 units; 4 or more for praevia, abruption, rupture or accreta); **HIV test** if unknown; **urine dipstick**; add **platelets, creatinine, LFTs and clotting** (or a bedside clotting test) for PE, abruption, sepsis or IUFD; **malaria RDT** if febrile. Full table and interpretation in section 6.

Decision and preparation (numbered checklist)

1. **Decide with a senior** (medical officer or obstetrician) and **assign the urgency category**. Write the indication clearly.
2. **Consent**: explain in her language (Luganda/Runyankole with an interpreter if needed) the indication, what will happen, the alternatives, and the risks. Risks include bleeding and transfusion, infection, injury to the bladder, bowel or ureter, VTE, hysterectomy (rare), anaesthetic risks, a cut to the baby (1–2%), and

implications for future pregnancies (rupture, praevia/accreta). Consent separately for **tubal ligation** or **PPIUD** if requested. She is the one who consents. The husband's signature is not legally required for an adult woman, although involving him is culturally wise.

3. **IV access**: two **large-bore (16–18 G)** cannulae. Take blood for **Hb and cross-match** (2 units; 4 for praevia, abruption or rupture).
4. **Fluids**: correct dehydration (e.g. obstructed labour) with **Ringer's lactate or normal saline**. Pre-load or co-load **500–1000 mL** crystalloid for spinal.
5. **Aspiration prophylaxis** (especially in labour): **metoclopramide 10 mg IV** plus **omeprazole 40 mg IV** (or an H2 blocker if available), and where available **sodium citrate 0.3 M 30 mL orally** just before induction of anaesthesia.
6. **Urinary catheter** (Foley 14–16 Fr) to empty the bladder, keep it out of the operative field and monitor urine output.
7. **Antibiotic prophylaxis within 60 minutes before skin incision** (drug table below).
8. **Vaginal cleansing with povidone-iodine** for women in labour or with ruptured membranes. This reduces post-op endometritis.
9. **Hair**: do **not shave**. Clip only if hair is in the way (shaving causes micro-abrasions that become infected).
10. In theatre: **left lateral tilt 15°**, **check the fetal heart again**, and (if she was fully dilated) **re-examine vaginally**.
11. **Neonatal preparation**: a person trained in **Helping Babies Breathe** with a bag-valve-mask, a warm towel and a resuscitaire at every CS.
12. **WHO Surgical Safety Checklist**: Sign in, Time out (the team confirms patient, procedure, category, antibiotics given, blood available), Sign out (counts, specimens, plan).

The pre-op sign-off before she leaves the ward

#	Item	Done ?
1	**Indication and urgency category** written in the notes and on the theatre request, with the **time of decision**	☐
2	**Consent** in her language (indication, procedure, risks, alternatives), separate consent for BTL/PPIUD	☐
3	**Anaesthetic review**: airway, fasting, platelets, clotting, anticoagulants, choice of spinal vs GA	☐
4	**Hb and group & cross-match** sent; blood in theatre if high risk	☐
5	Two **16–18 G cannulae**; fluids running if dehydrated	☐
6	**Foley catheter** in, urine colour noted	☐
7	**Antacid prophylaxis** (metoclopramide + omeprazole; sodium citrate if available)	☐
8	**Antibiotic prophylaxis** drawn up to give **within 60 minutes before incision** (therapeutic course if she is already infected)	☐
9	**HIV status** known; eMTCT drugs ready for the baby	☐
10	**Neonatal resuscitator** called	☐
11	**WHO Surgical Safety Checklist** (Sign in, **Time out**, Sign out), including a **repeat FHR check** and **repeat VE** if fully dilated	☐

Anaesthesia

- **Spinal (subarachnoid) anaesthesia is the method of choice** for most CS. The mother is awake, there is **no airway risk** (failed intubation and aspiration were historically major causes of anaesthetic death in pregnancy), there is less blood loss, and she has early contact with the baby.

- **Contraindications to spinal**: section 2.
- **General anaesthesia (GA)** is used when a spinal is contraindicated or too slow (some Category 1 cases), or when it fails. It needs rapid sequence induction with cricoid pressure, and a skilled anaesthetist.
- **Spinal complications**: hypotension (the commonest; tilt, fluids, vasopressor), high or total spinal (breathlessness, arm weakness, bradycardia, apnoea), nausea, shivering, **post-dural puncture headache** (worse when upright; fluids, analgesia, caffeine; blood patch if severe), and rare meningitis.

Key peri-operative drugs

Drug	Dose	Route	Frequency / timing	Notes
Cefazolin (WHO/ACOG first choice)	**2 g** (3 g if ≥120 kg)	IV	Single dose, **15–60 min before incision**	First-generation cephalosporin; narrow; ideal
Ampicillin (alternative)	**2 g**	IV	Single dose before incision	Cheap; widely available
Ceftriaxone (common Ugandan practice)	**1–2 g**	IV	Single dose before incision	Broad spectrum. Rising resistance in Uganda, so reserve prolonged use for real infection
Metronidazole (often added in labour/ROM)	**500 mg**	IV	Single dose pre-op	Anaerobic cover
Azithromycin (ACOG adjunct for CS in labour/ROM)	500 mg	IV	Single dose	Reduces endometritis; availability is limited in Uganda
Clindamycin + gentamicin (severe penicillin allergy)	900 mg + 5 mg/kg	IV	Single dose	
Metoclopramide + omeprazole (aspiration prophylaxis)	10 mg + 40 mg	IV	Pre-op	Sodium citrate 0.3 M 30 mL PO just before induction where available
Oxytocin	**10 IU** (WHO); UK practice is **5 IU**	Slow IV (or IM)	After delivery of the baby	Then an infusion: **20 IU in 1 L** NS/RL at about 40 drops/min for 4–8 h in high-risk women
Carbetocin (if available)	100 µg	Slow IV	Single dose	Heat-stable; NICE preferred at CS
Tranexamic acid	**1 g** over 10 min	IV	Within 3 h of birth for PPH; may repeat 1 g after 30 min	Treatment of PPH (WOMAN trial). Some units give it prophylactically for high-risk CS
Hyperbaric bupivacaine 0.5% (spinal)	**2–2.5 mL (10–12.5 mg)** ± fentanyl 10–25 µg	Intrathecal (L3/4 or L4/5)	Once	Given by the anaesthetist/anaesthetic officer
Phenylephrine (or **ephedrine**) for spinal hypotension	Phenylephrine 50–100 µg bolus; ephedrine 6 mg bolus	IV	Repeat to keep SBP near baseline	Hypotension harms the baby, so treat early

Drug	Dose	Route	Frequency / timing	Notes
Enoxaparin (thromboprophylaxis)	**40 mg** (50–90 kg); 20 mg if <50 kg; 60 mg if 91–130 kg	SC	Once daily, from 4–6 h post-op (≥4 h after spinal)	Emergency CS means at least **10 days**; very high risk means **6 weeks** (RCOG GTG 37a). Alternative: **unfractionated heparin 5000 IU SC 12-hourly**

DRUG DOSING

Prophylaxis is **one dose**. Repeat the dose only if the operation lasts more than about 3–4 hours or blood loss exceeds 1500 mL. Prolonged "prophylactic" courses (for example ceftriaxone for 5 days after every CS) do **not** prevent SSI. They breed resistance and are recommended against by the WHO SSI guideline. Give a **therapeutic** course only when there is established infection (chorioamnionitis, prolonged obstructed labour with fever, gross contamination).

UGANDA CONTEXT

Cefazolin is not stocked in many Ugandan public hospitals. The usual practice (Mulago/Kawempe, Mengo, most RRHs) is **ceftriaxone 1–2 g IV ± metronidazole 500 mg IV given at induction of anaesthesia**, i.e. before incision. A Mulago RCT showed that giving it **before incision** rather than after cord clamping reduced post-op infection. Follow your unit protocol, but insist on the **timing**. Where enoxaparin is unavailable, give **UFH 5000 IU SC 12-hourly**, and in every case use **early mobilisation and hydration**.

5. Procedure Step by Step

A. Surgical steps of a lower-segment CS

1. **Position** supine with 15° left tilt. Catheter in. **Skin prep** with chlorhexidine-alcohol (or povidone-iodine), letting it **dry** (it is flammable with diathermy). Drape.
2. **Skin incision**:
   - **Joel-Cohen**: straight transverse, about **3 cm above the symphysis**, higher than Pfannenstiel. The later layers are opened mostly **bluntly**. It is faster, with less blood loss, fever and pain, and is recommended by NICE.
   - **Pfannenstiel**: curved (smile-shaped) transverse incision **2 fingerbreadths (about 2–3 cm) above the symphysis**, 12–15 cm long, with sharp dissection. It gives a good cosmetic result.
   - **Subumbilical midline**: when **rapid wide access** is needed (suspected **ruptured uterus**, a huge fibroid, anticipated hysterectomy, previous midline scar), or sometimes for a Category 1 CS by less experienced surgeons.
3. **Fat**: incise centrally, then separate with the fingers.
4. **Rectus sheath**: small transverse incision, extended laterally with **Mayo scissors** or bluntly. In a Pfannenstiel, the sheath is lifted with **Kocher's** forceps and freed from the rectus muscles. **Separate the recti** in the midline with the fingers.
5. **Parietal peritoneum**: open **high** (bladder!) with fingers, or lift it between two artery forceps, feel that nothing is caught, and incise. Extend with care.
6. **Doyen retractor** over the bladder. Check for **dextrorotation** by checking that the round ligaments are equidistant, so the incision is central and doesn't run into the left uterine vessels.
7. **Uterovesical fold**: incise transversely and push the bladder down (a bladder flap). Omitting the flap is acceptable in most cases.

8. **Uterine incision**: a transverse 2–3 cm incision in the lower segment with the scalpel, **stroke by stroke** (to avoid cutting the baby), until membranes or liquor appear. Then **extend bluntly** with the fingers laterally and slightly upwards. Blunt extension has less blood loss and fewer uncontrolled extensions than cutting with scissors.
9. **Delivery**: remove the Doyen. Slip a hand below the head, flex it and lift it out of the incision while the assistant gives **fundal pressure**. Suction the mouth and nose only if needed. Deliver the body. **Delay cord clamping for at least 1 minute** in a vigorous baby.
    - A **deeply impacted head** (second-stage CS): the assistant pushes the head up vaginally ("push"), or deliver by the feet first ("pull", reverse breech). Call a senior early.
    - **Breech or transverse**: grasp the feet (breech extraction); enlarge the incision if needed (J or inverted-T, which must be documented because it changes all future births).
10. **Oxytocin 10 IU slow IV** (or IM). **Placenta by controlled cord traction** rather than manual removal (less endometritis and blood loss). **Check the cavity is empty** with a dry swab.
11. **Uterine closure**: **Green-Armytage** clamps on the angles and bleeding edges. Close in **two layers** with **#1 absorbable (polyglactin/Vicryl or chromic catgut)**: a continuous (often locked) first layer, then an imbricating second layer. Two-layer closure is standard in Uganda and recommended by NICE, and it may reduce later rupture. **Secure the angles** beyond the apex, where most bleeding comes from.
12. **Check haemostasis**, tubes and ovaries. Perform **tubal ligation** if consented (e.g. modified Pomeroy with a Babcock).
13. **Peritoneal toilet**: remove clots and liquor from the gutters. Do the **swab, needle and instrument count**.
14. **Peritoneum**: NICE advises that visceral and parietal peritoneum need not be closed. Many Ugandan surgeons close the parietal layer; both approaches are acceptable.
15. **Rectus sheath**: continuous **#1 Vicryl (or PDS/nylon loop)**, bites about 1 cm from the edge and 1 cm apart, not strangulating.
16. **Subcutaneous fat**: close only if **>2 cm thick** (reduces wound separation).
17. **Skin**: subcuticular absorbable (3-0 Vicryl/Monocryl), interrupted nylon, or staples.
18. **Finish**: fundal massage and **vaginal toilet** (express clots), check the uterus is contracted, **estimate blood loss**, and write the **operation note** (section 9).

B. When is a classical CS indicated?

Indication	Why
Very preterm (<28–30 wk) with an undeveloped lower segment, especially with breech/transverse	No room for a safe transverse lower-segment incision
Transverse lie, back down, with ruptured membranes (arm prolapse, impacted)	The fetus cannot be reached through the lower segment
Anterior placenta praevia with large vessels / accreta	To avoid cutting through the placenta; may precede hysterectomy
Large lower-segment fibroid or dense adhesions/bladder over the lower segment	Lower segment inaccessible
Invasive cervical cancer	Avoid cutting into the tumour; may proceed to radical surgery
Perimortem CS (at 4–5 min of maternal arrest)	Speed

EXAM FAVOURITE

Classical scar implications: rupture risk in later pregnancy is **4–9%**, and rupture can occur **before labour** (in the third trimester). So every future birth must be by **elective CS at 36–37 weeks, before labour**, and she must never labour at home. Document this in bold on the discharge form and her Mother–Child Health Passport.

C. Immediate post-operative care (recovery room, first 2 hours)

1. Vital signs (**pulse, BP, RR, SpO_2**) every **15 minutes for the first hour**, then every 30 minutes for 2 hours, then 4-hourly.
2. **Uterine tone and lochia** every 15 minutes. **Rub up** the uterus if it is soft and look for PPH.
3. Wound dressing (soakage), **urine output** (≥30 mL/h), level of spinal block (recovering legs).
4. **Skin-to-skin and breastfeeding within the first hour.**
5. **Oxytocin infusion 20 IU in 1 L at about 40 drops/min** for high-risk women (prolonged labour, twins, polyhydramnios, praevia, PPH at surgery).

Analgesia (multimodal, regular, not "PRN only")

Drug	Dose	Route	Frequency	Notes
Paracetamol	1 g	Oral / IV	6-hourly (max 4 g/day)	Baseline for everyone
Diclofenac	50 mg oral 8-hourly, or 75 mg IM 12-hourly; or 100 mg rectal	Oral / IM / PR	For 48–72 h	Avoid in **severe PE with renal impairment**, asthma, PPH with oliguria, peptic ulcer. Safe in breastfeeding
Tramadol	50–100 mg	Oral / IM / IV	6–8-hourly PRN	Moderate pain
Morphine	5–10 mg (0.1 mg/kg)	IM (or titrated IV)	4-hourly PRN	Severe pain first 24 h; monitor RR and sedation
Pethidine (still used in Uganda)	50–100 mg	IM	4-hourly PRN	Short-acting; nausea

D. Ward care and ERAS (Enhanced Recovery After Surgery)

- **Eat and drink early**: sips of water within 1–2 hours and light food when hungry, if uncomplicated. The old rule of "nil by mouth until bowel sounds or flatus" is **not** needed after an uncomplicated CS and prolongs IV fluids, hunger and ileus.
- **Early mobilisation**: sit out and walk within **6–12 hours**. This prevents VTE, chest infection and ileus.
- **Catheter out** once she is mobile, **≥12 hours after the last spinal dose** (usually the next morning). Keep it **7–14 days** if the **bladder was injured and repaired**, or after prolonged obstructed labour with haematuria (to prevent a VVF from pressure necrosis).
- **Stop IV fluids** once she is drinking.
- **Thromboprophylaxis**: risk-assess every woman. Give LMWH or UFH for emergency CS in labour and any extra risk factors (obesity, age >35, PE, infection, PPH >1 L, immobility, sickle cell).
- **Wound**: remove the dressing at **24–48 h** and keep it clean and dry. Remove non-absorbable skin sutures on **day 5–7** (Pfannenstiel day 5; midline 7–10).
- **Antibiotics**: none after surgery unless there is infection.

- **Daily review**: the day-1 "head to toe, then the 6 Ps" review (section 6) and the questions: pain control; nausea or vomiting; **eating and drinking**; **passing flatus**; **passing urine** after the catheter is out; **mobilising**; **breastfeeding** and the baby's condition; **headache** (post-dural puncture headache is worse on standing; severe headache with high BP means pre-eclampsia); **calf pain or breathlessness** (VTE); **fever, rigors, foul lochia or wound discharge** (infection); **heavy bleeding**.
- **Discharge**: usually on day 2–3 if afebrile, eating, mobile, voiding, the wound is clean, Hb is adequate (give **ferrous sulphate 200 mg + folic acid** for 3 months if anaemic), and breastfeeding is established.

E. Discharge counselling: before discharge, every woman should know

1. **Why** she had a CS, and whether the reason is likely to recur.
2. **What type of uterine incision** she has. Write it on her card.
3. The **plan for the next birth**: a hospital (HC IV or above) delivery; TOLAC versus ERCS; never deliver at home or at an HC II/III.
4. **Spacing**: wait **at least 18–24 months before the next conception** (WHO recommends a ≥24-month interpregnancy interval). Offer contraception: **PPIUD** (can be placed at CS), implant, injectable, lactational amenorrhoea, or BTL.
5. **Danger signs**: fever, foul lochia, heavy bleeding, wound discharge or opening, calf pain, breathlessness, severe headache, and not passing urine.
6. Postnatal review at **6 days and 6 weeks**. HIV-exposed baby: NVP prophylaxis and an EID plan.

6. Findings and Interpretation

Bedside tests

Test	Before CS	After CS
HemoCue / point-of-care Hb	Baseline; plan blood	Day 1 if EBL was high, she is pale or tachycardic
Urine dipstick	**Protein** (pre-eclampsia), **blood** (obstruction, bladder injury), **ketones** (dehydration after long labour)	Nitrites and leucocytes if urinary symptoms or fever
RBS	Diabetes, or a woman on steroids	If she is diabetic
Bedside clotting test	Abruption, severe PE/HELLP, IUFD, sepsis, or massive bleeding	Ongoing bleeding
Malaria RDT	Any fever	Any fever (malaria is common in Uganda)
HIV rapid test	If status unknown, or no negative test late in this pregnancy (follow the national retesting schedule)	
Pinard / Doppler FHR	On decision and **again in theatre before incision**	
Point-of-care ultrasound (if skilled)	Viability, presentation, **placental site**	Free fluid, a large clot in the uterus, wound collection
Partograph review	Has she crossed the action line? Descent and moulding trend?	

CLINICAL PEARL

Bedside clotting test (WHO MCPC): put **2 mL of venous blood** in a small, clean, dry plain **glass** tube and hold it in your closed fist to keep it warm. After **4 minutes** tip it gently, then every minute. If **no clot has formed by 7 minutes**, or a soft clot breaks down easily, she has a **coagulopathy**. Tell the anaesthetist (spinal is contraindicated) and get fresh whole blood or FFP ready. (Some units use the 20-minute whole-blood clotting test instead: no clot at 20 minutes = coagulopathy.)

Laboratory and imaging

Test	When	Expected / abnormal finding	How it changes management
Hb / FBC (HemoCue at HC IV; full FBC at hospital)	Every CS, before surgery; day 1–2 after if EBL ≥1 L or symptomatic	Hb ≥11 g/dL normal; **<7 g/dL severe**. WBC up to ~25 × 10^9/L can be normal in labour. **Platelets <75–80 × 10^9/L**	Transfusion planning; low platelets rule out a spinal; post-op anaemia means iron (or transfusion if symptomatic)
Blood group and cross-match (HC IV with a blood bank; always at RRH/national)	Every emergency CS	**2 units** routinely; **4+ units** for praevia, abruption, rupture or suspected accreta	Blood must be in theatre before a high-risk CS starts
HIV rapid test (all levels)	Status unknown, or no recent negative test (national retesting schedule)	Reactive → confirm per national algorithm	Same-day ART for the mother; NVP prophylaxis and EID plan for the baby; double gloving
Urinalysis (dipstick at all levels)	Every woman	Protein; blood; ketones	Suggests PE, obstructed labour or dehydration
Random blood sugar (all levels)	Diabetes, steroids, obesity	>11 mmol/L is high	Peri-op glucose control (reduces SSI)
Creatinine, electrolytes, LFTs (RRH/national; some HC IVs)	Severe PE, HELLP, abruption, sepsis, prolonged obstructed labour	Creatinine >90 µmol/L, raised ALT/AST	Avoid NSAIDs; fluid plan; HDU care
Clotting: PT, APTT, fibrinogen (RRH/national; bedside clotting test elsewhere)	Abruption, HELLP, IUFD, massive PPH	**Fibrinogen <2 g/L** = severe coagulopathy	No spinal; FFP, cryoprecipitate or fresh whole blood
Sickle cell test (where indicated)	Known or suspected SCD	HbSS	Hydration, oxygen, warmth, transfusion plan, thromboprophylaxis
Malaria RDT or blood smear (all levels)	Any fever before or after CS	Positive	Treat malaria; fever in labour is not always chorioamnionitis
Blood cultures (RRH/national; often unavailable at HC IV)	Fever with sepsis signs, before or after CS	Growth of Gram-negatives (often ESBL) or *Staph*	Switch from empirical to targeted antibiotics
Pus swab / wound culture (RRH/national)	Discharging or opened wound	*Klebsiella*, *E. coli* (often **ceftriaxone-resistant**), MRSA	Guides antibiotics; alert IPC if several cases
Urine culture (RRH/national)	Fever with urinary symptoms, or after a long catheter	Significant growth	Targeted antibiotic
Obstetric ultrasound (many HC IVs have a machine; skill varies)	Before an elective or planned CS; APH; suspected IUFD	Viability, presentation, **placental site** (praevia, accreta signs), EFW	Plans incision, blood and seniority; confirms IUFD (vaginal birth usually better)
Pelvic/abdominal ultrasound after CS (RRH/national)	Fever not settling after 48–72 h of antibiotics, a mass or wound swelling	Pelvic, subsheath or wound **collection**, retained products	Drainage or re-laparotomy
Doppler ultrasound of the leg / CTPA (national referral or private)	Suspected DVT or PE	Thrombus	Therapeutic LMWH; do not wait for imaging if PE is likely

Three points to remember:

- **Blood is the investigation that saves lives at CS.** Send the cross-match at the time of decision, not when the bleeding starts: **2 units** for a routine emergency CS and **4 or more** for praevia, abruption, rupture or accreta.
- **Platelets and clotting decide the anaesthetic.** A spinal is safe with platelets above about **75–80 × 10^9/L** and a normal bedside clotting test.
- **Fever is not always chorioamnionitis or a wound infection in Uganda.** Always do a malaria RDT or smear.

The day-1 post-CS ward review ("head to toe, then the 6 Ps")

Check	Normal	Abnormal: think
Pulse, BP, temperature, RR	Pulse <100, afebrile	Tachycardia: bleeding, sepsis, PE, anaemia. **Fever ≥38.0 °C**: see the fever workup (section 8)
Pallor	Pink conjunctivae	Anaemia: check Hb
Chest	Clear	Basal crackles (atelectasis, aspiration)
Breasts	Soft, milk coming	Engorgement, cracked nipples
Fundus	Firm, at or below the umbilicus, mildly tender	**Soft, rising** (PPH, clots); **very tender** (endometritis)
Abdomen	Soft, mildly distended, bowel sounds by day 1–2	**Tense, silent, distended** (ileus, haemoperitoneum, bowel injury)
Wound (dressing off at 24–48 h)	Dry, edges together	Soaked dressing (haematoma), redness, induration, pus, gaping
Lochia (pad)	Red, moderate, no smell	Heavy (PPH); **offensive** (endometritis)
Urine	**≥30 mL/h**, clear	Oliguria (hypovolaemia, PE); **haematuria** (bladder injury, obstructed labour)
Calves	Soft, non-tender	Swollen, tender (DVT)

Interpreting deterioration in labour in a woman with a scar (or in obstructed labour)

Condition	Distinguishing features	Key investigation / action
Uterine rupture	Scar or obstruction, **abnormal FHR (earliest)**, constant pain, contractions stop, loss of station, easily felt fetal parts, shock out of proportion to visible bleeding, haematuria	Clinical diagnosis. **Immediate laparotomy.** Bedside US may show free fluid or a fetus outside the uterus, but don't delay
Placental abruption	Painful dark bleeding, **woody-hard tender uterus**, contractions continue, fetal distress or IUFD, shock, may have PE	Clinical; Hb, clotting (bedside clotting time) for DIC; deliver
Placenta praevia	**Painless** bright bleeding, **soft non-tender** uterus, high head or malpresentation, FHR usually normal	Ultrasound for placental site. **No digital VE**
Obstructed labour without rupture	Prolonged labour, 3+ moulding, caput, head ≥3/5, Bandl's ring, oedematous vulva, maternal exhaustion, haematuria	Clinical and partograph (past the action line). CS
Chorioamnionitis	Fever, maternal and fetal tachycardia, uterine tenderness, foul liquor	FBC, blood culture; therapeutic antibiotics; deliver
Scar tenderness only / "scar pain"	Local tenderness with normal FHR, vitals and progress	Close monitoring. A normal FHR is reassuring

Worked example 1: interpreting pre-op results

A woman with a **placental abruption** and a live baby at 36 weeks: Hb 8.9 g/dL, platelets 62 × 10^9/L, bedside clotting test **no clot at 7 minutes**. *Interpretation:* anaemia plus **consumptive coagulopathy (DIC)**. *What changes:* a **spinal is contraindicated** (low platelets and coagulopathy), so she needs **GA** with an experienced anaesthetist; cross-match **4 units** and ask for **fresh whole blood or FFP**; give **tranexamic acid** if bleeding at surgery; prepare for **PPH** (oxytocin infusion, uterine compression sutures); and plan HDU care afterwards.

7. Clinical Acumen

First 5 minutes: when you are called

To a woman who may need a CS:

1. **Look at her and at the baby's heart first.** Airway and breathing, colour, conscious level. Take **pulse, BP, RR, temperature and SpO_2**, and work out the **shock index** (HR ÷ SBP; **≥0.9 is abnormal**, **≥1.4 needs urgent intervention**). Listen to the **fetal heart for a full minute after a contraction** (normal 110–160 bpm).
2. **Look for the Category 1 situations**: cord prolapse, sustained fetal bradycardia, suspected uterine rupture, major abruption with a live baby, maternal collapse. These cannot wait for a full clerking.
3. **Call for help early**: the medical officer or obstetrician (to confirm the decision), the **anaesthetist/anaesthetic officer**, the theatre team, and the lab for blood. **Say the category out loud.**
4. **Start in parallel** (one person each, not one after another): two **16–18 G cannulae**, bloods for **Hb and cross-match**, **stop any oxytocin** if the FHR is abnormal or rupture is suspected, **left lateral position**, oxygen if she is shocked, a **Foley catheter**, and fluids (Ringer's lactate or normal saline) if she is dehydrated or bleeding.
5. **Stabilise first when the mother is the patient** (section 2).

To a woman after a CS (bleeding, low BP, fever, "not looking well"): pulse, BP, RR, SpO_2, temperature, **fundus (height and tone)**, **lochia and pad**, **wound dressing**, **urine output** in the catheter bag, and a quick look at the drug and fluid chart. A **pulse >100 with a rising fundus or a soaked pad** is PPH until proven otherwise. Rub up the uterus and call your senior.

Red flags: escalate now

RED FLAG

- **Fetal bradycardia (<110 bpm sustained) or recurrent decelerations in a woman with a scar** means suspect rupture. Category 1.
- **Maternal tachycardia, hypotension, pallor or collapse** in labour means haemorrhage (abruption, rupture) until proven otherwise.
- **Constant abdominal pain, shoulder-tip pain, contractions that suddenly stop, presenting part that recedes (loss of station), fetal parts felt easily under the skin**: uterine rupture.
- **Bandl's ring, haematuria, oedematous vulva, 3+ moulding, head ≥3/5 palpable** after prolonged labour: obstructed labour. Do not augment; prepare CS.
- **Cord felt in the vagina** (cord prolapse): Category 1 if the baby is alive and she is not fully dilated.
- **Fever >38 °C, foul liquor, fetal tachycardia**: chorioamnionitis. Give therapeutic antibiotics, not prophylaxis only.
- **Post-op**: pulse >100, falling BP, rising fundus, heavy lochia, a distended silent abdomen, fever after day 2, wound discharge, calf swelling, breathlessness.

Thinking like a clinician

1. **Is vaginal birth impossible, or just harder?** Absolute indications (major praevia, transverse lie in labour, true CPD, previous classical scar) have no alternative. For relative indications, weigh the risk to the baby now against the risk to this woman now and in every future pregnancy.
2. **Is the baby alive?** A CS for a dead baby exposes the mother to all the risk for none of the benefit (section 2).
3. **"If X then think Y"**:
   - Previous CS plus a **new** abnormal FHR: **rupture**, not "fetal distress".
   - Prolonged labour plus **haematuria** plus a high head: **obstructed labour** with bladder compression. Expect a thin lower segment and a high bladder at surgery, so incise the peritoneum high.
   - Fully dilated plus a **deeply impacted head**: expect a difficult delivery of the head and extensions. Call a senior. Prepare the "push" (an assistant pushes up from the vagina) or "pull" (reverse breech extraction) technique.
   - APH plus a soft non-tender uterus plus a high or malpresenting part: **placenta praevia**. **No VE outside theatre.**
   - Fever before surgery: take cultures, then give **therapeutic** antibiotics (not a single dose).
4. **Common student pitfalls:**
   - Assuming a Pfannenstiel skin scar means a lower-segment uterine incision. Not necessarily.
   - Forgetting to check the FHR in theatre, or forgetting to re-examine a fully dilated woman (a ventouse may be possible).
   - Writing "fetal distress" without describing the FHR (rate, pattern, relation to contractions, duration).
   - Giving antibiotics "after the baby is out".
   - Keeping women "nil by mouth until bowel sounds return". This is an outdated practice.
   - Forgetting the **next-pregnancy counselling** at discharge.

Planning the next birth: trial of labour after CS (TOLAC) / VBAC

Who is a good candidate?

- **One** previous **lower-segment transverse** CS with an uncomplicated recovery.
- Singleton, **cephalic**, term (37–41 weeks), **spontaneous onset** of labour.
- **Non-recurrent** previous indication (breech, fetal distress, APH); a **previous vaginal birth** helps.
- Inter-delivery interval **≥18 months**; no other indication for CS; a clinically adequate pelvis; EFW under about 4 kg.
- **Delivering in a facility that can perform an immediate CS and transfuse (HC IV/hospital) with continuous midwife presence.**

Contraindications to TOLAC: previous **classical, T or J incision**; previous **uterine rupture**; **placenta praevia**; any other indication for CS (e.g. transverse lie); **2 or more previous CS** (in most Ugandan units; RCOG allows a carefully counselled trial); **a facility without CS capability**.

Counselling numbers (RCOG GTG 45)

	Planned VBAC	Elective repeat CS
Chance of vaginal birth	**72–75%** (85–90% if previous vaginal birth)	0%
Uterine rupture	**About 0.5% (1 in 200)**; 2–3 times higher with induction or augmentation	Extremely rare (about 0.02%)
Advantages	Shorter recovery, avoids surgery, lower risk of praevia/accreta and surgical complications in future pregnancies	Planned, avoids labour and rupture

	Planned VBAC	Elective repeat CS
Disadvantages	About 1 in 4 need an emergency CS in labour (higher morbidity than an elective CS)	Surgical risks; each additional scar raises future accreta risk; neonatal TTN

Management of TOLAC in labour ("trial of scar")

1. Admit early in labour. Confirm the previous incision type from the notes. Take **Hb and group & save/cross-match**, and insert an IV cannula.
2. Monitor with a **partograph**. FHR: continuous CTG where available (RCOG). In most Ugandan units, **intermittent auscultation every 15 minutes in the first stage and every 5 minutes in the second stage**, always after a contraction.
3. Watch maternal pulse, pain and scar tenderness, vaginal bleeding and haematuria, descent (fifths) and contractions.
4. **Progress**: expect normal progress. **Do not allow the partograph to cross the action line**; slow progress means reassessment by a senior and usually a CS.
5. **Oxytocin augmentation**: only with a senior decision, in a hospital, at a low dose with careful titration. Many Ugandan units avoid it.
6. **Induction**: **misoprostol is contraindicated** with a uterine scar at term. Use **Foley catheter ripening, then ARM, then cautious oxytocin** only at a referral hospital, with counselling about the higher rupture risk.
7. Second stage: avoid prolonged pushing. Consider an assisted delivery (ventouse) for delay. Some units routinely shorten the second stage.
8. After a VBAC, watch for PPH and signs of scar dehiscence (unexplained tachycardia, pain, bleeding, haematuria).

What happens at HC IV vs referral hospital

Setting	What to do
HC III	Identify women with a scar or CS indication at ANC and refer them for **delivery at HC IV/hospital**. In labour: partograph; refer early if progress is slow or the scar is tender. Before referral: IV line, fluids, catheter if obstructed, **first-dose antibiotics** if ROM is prolonged or she has a fever, and a referral note. Send with a health worker and a blood donor
HC IV (CEmONC)	Routine and emergency LSCS under spinal, transfusion, TOLAC for well-selected women, repair of simple rupture, bladder repair only if skilled
General / regional referral hospital	Complex CS: 2 or more previous scars, praevia/accreta, second-stage impacted heads, rupture needing hysterectomy, severe PE/HELLP, cardiac disease, bladder/ureteric injury, ICU/HDU care, VBAC induction with Foley
National referral (Kawempe/ Mulago)	Placenta accreta spectrum, multiple scars with complications, fistula management, ICU

8. Complications and Troubleshooting

Timing	Maternal	Neonatal
Intra-operative	**Haemorrhage** (atony, angle extensions into the broad ligament and uterine vessels, praevia/accreta); **bladder injury** (repeat CS, obstructed labour); ureteric injury (lateral extensions, blind clamping); bowel injury (adhesions); anaesthetic (hypotension, high spinal, failed intubation, aspiration); amniotic fluid embolism	**Fetal laceration** (1–2%); difficult delivery of an impacted head causing skull fracture or hypoxia

Timing	Maternal	Neonatal
Early post-op (days)	**PPH**; **endometritis and SSI** (the commonest); UTI; **wound haematoma, dehiscence or burst abdomen**; paralytic ileus; **VTE** (DVT/PE); urinary retention; post-dural puncture headache; chest infection; **sepsis and death**	**Transient tachypnoea of the newborn** (especially elective CS before 39 weeks); respiratory distress; delayed breastfeeding
Late	Adhesions and chronic pelvic pain; incisional hernia; **caesarean scar niche** (isthmocele) causing spotting, dysmenorrhoea and subfertility; **scar ectopic pregnancy**; **uterine rupture** in a later pregnancy; **placenta praevia and placenta accreta spectrum** (the risk climbs with each CS)	Possibly altered microbiome and a small increase in asthma or obesity (association)

CLINICAL PEARL

Placenta accreta risk climbs steeply with each CS, especially when the placenta is praevia over the scar. With a praevia, the accreta risk is about **11% after one previous CS, about 40% after two and over 60% after three** (Silver et al., 2006, a large US cohort). This is the strongest argument for avoiding the *first* unnecessary CS in a woman who wants a large family.

Troubleshooting

Problem	Action
Spinal hypotension	Left tilt, IV fluids, phenylephrine 50–100 µg or ephedrine 6 mg bolus; treat early (hypotension harms the baby)
High or total spinal	Breathlessness, arm weakness, bradycardia, apnoea: airway support and ventilation by the anaesthetist
Post-dural puncture headache	Worse upright: fluids, analgesia, caffeine; blood patch if severe
Deeply impacted head	Call a senior; "push" from below or "pull" (reverse breech)
Lateral extension / angle bleeding	Green-Armytage clamps, secure the angles beyond the apex; identify the ureter before deep sutures
Atony / PPH at surgery	Oxytocin infusion, tranexamic acid 1 g IV, uterine compression sutures; hysterectomy if needed
Bladder injury	Repair in 2 layers; catheter for **7–14 days**
Post-op PPH (pulse >100, rising fundus, soaked pad)	Rub up the uterus, oxytocin, call the senior; return to theatre if bleeding continues
Wound discharge	Open the wound at the point of discharge, drain pus, swab for culture, daily dressing
Fever not settling after 48–72 h of antibiotics	Ultrasound for a pelvic or subsheath collection or retained products; consider a resistant organism; drain

Worked example 2: the post-CS fever workup

A woman is febrile (**38.6 °C**, pulse 108) on **day 4** after an emergency CS for obstructed labour. Think through the causes by timing, using the **"W" list**:

Cause	Typical day	Clues	Test
Womb (endometritis)	Day 2–5	Tender uterus, **offensive lochia**, long labour, many VEs	FBC, blood culture; high vaginal swab
Wound (SSI)	**Day 4–7**	Redness, induration, pus, pain out of proportion	Pus swab culture before antibiotics

Cause	Typical day	Clues	Test
Water (UTI)	Day 3–5	Catheter, dysuria, loin pain	Dipstick, urine culture
Wind (chest)	Day 1–2	Cough, crackles, GA or aspiration	SpO_2; chest X-ray if available
Walking (VTE)	Day 5 onwards	Calf swelling, breathlessness, tachycardia	Doppler; treat if likely
Weaning (breasts)	Day 3 onwards	Engorged, red, tender breast	Clinical
Wonder drugs / drip	Any	Cannula-site phlebitis, drug rash	Look at every cannula site
Malaria (Uganda)	Any	Rigors, no other focus	**RDT or smear**

On examination she has **offensive lochia, a tender bulky uterus** and a **red, indurated wound with a small pus leak**. *Interpretation:* **endometritis plus a superficial incisional SSI**. *Plan:* sepsis screen (FBC, blood cultures, pus swab, RDT), then **IV ampicillin 2 g 6-hourly + gentamicin 5 mg/kg daily + metronidazole 500 mg 8-hourly** (or the unit's ceftriaxone + metronidazole), **open the wound at the point of discharge** (remove 1–2 skin sutures) to drain pus, daily dressing, fluids and paracetamol, and **review the culture**. If there is no improvement after **48–72 h**, look for a **pelvic or subsheath collection** (ultrasound) or **retained products**, and ask whether the organism is resistant. If several women on the ward have similar infections, **tell the IPC focal person** (see the IPC topic).

> EXAM FAVOURITE
> **SSI classification (CDC/WHO surveillance):** infection **within 30 days** of surgery. **Superficial incisional** = skin and fat; **deep incisional** = fascia and muscle; **organ/space** = uterus (endometritis), pelvic abscess. A deep or organ/space SSI with a fever that will not settle usually needs **drainage**, not just "stronger antibiotics".

Emergency: suspected uterine rupture (algorithm)

1. **Call for help**: senior obstetrician, anaesthetist, theatre team, lab for blood. **Category 1 CS / laparotomy.**
2. **Stop oxytocin.** Position left lateral. Give oxygen 10–15 L/min by face mask.
3. **Two 16 G cannulae**; send Hb and **cross-match 4 units**; bedside clotting test.
4. **Fluids**: warm crystalloid (RL/NS) 1–2 L rapidly, then **blood** as soon as available. Aim for SBP ≥90 and urine output ≥30 mL/h. Avoid over-infusing crystalloid; permissive resuscitation until the bleeding is controlled.
5. **Catheterise** (haematuria suggests bladder involvement).
6. **Antibiotics**: ampicillin 2 g IV + metronidazole 500 mg IV + gentamicin 5 mg/kg IV (or ceftriaxone 2 g IV + metronidazole), then continue as a treatment course.
7. **Laparotomy through a midline incision**: deliver the baby and placenta, find the bleeding points, and clamp.
8. **Repair vs hysterectomy**:
   - **Repair** (two layers) if the tear is clean, not too extensive, she is stable, and she wants more children. Strongly offer **tubal ligation** or counsel that any future pregnancy needs an elective CS at 36–37 weeks.
   - **Subtotal (or total) hysterectomy** if the tear is extensive, involves the uterine vessels or cervix, the tissue is infected or necrotic, or she is unstable. Life comes before fertility.
9. **Inspect the bladder** (repair in 2 layers, catheter for 10–14 days) and the ureters.
10. Post-op: HDU-level monitoring, transfusion, antibiotics, thromboprophylaxis once bleeding has stopped, counselling and bereavement support, and **an MPDSR review** for a maternal or perinatal death or near miss.

9. Documentation and Presenting Findings

Model pre-op summary line: "Mrs B.K., 26, G2P1 at 39+4 weeks with **one previous LSCS for breech**, in labour for 14 h, now **8 cm for 4 hours with the head 3/5 palpable, moulding +3 and haematuria**, FHR 150. Impression: **obstructed labour with a scarred uterus, no signs of rupture yet**. Plan: **Category 2 emergency LSCS**."

Problem list: (1) obstructed labour; (2) previous CS (rupture risk); (3) haematuria (bladder at risk, long catheter afterwards); (4) dehydration and ketosis; (5) infection risk (long labour, many VEs).

Documenting the decision: indication in one line, absolute or relative, urgency category, **time of decision**, the FHR described (rate, pattern, relation to contractions, duration), not just "fetal distress".

The operation note must include: indication, category, times, anaesthesia, **incision type (skin and uterus)**, findings, complications, **number of layers of uterine closure**, EBL, counts, and the **plan for the next pregnancy**. A J or inverted-T extension must be documented because it changes all future births.

Discharge documentation: the uterine incision type and the next-birth plan on her card and **Mother–Child Health Passport** (a classical scar in bold: elective CS at 36–37 weeks, never labour at home).

Model post-operative presentation

> "Mrs A.N., 24-year-old **P2+0**, day **1** after an **emergency lower-segment CS**, **Category 2**, done yesterday at 18:00 under **spinal** for **obstructed labour** (Robson group 3). Intra-operatively the lower segment was thin, the bladder was high but intact, there was a live male baby weighing 3.6 kg with Apgars 7 and 9, the uterus was closed in 2 layers, and EBL was 700 mL. She received ceftriaxone 2 g and metronidazole 500 mg IV at induction. This morning she has mild wound pain, is tolerating oral fluids, and has not yet mobilised. Pulse 88, BP 118/72, temp 37.2 °C, not pale, uterus well contracted at the umbilicus, lochia normal, wound dry, calves soft, and urine clear at 60 mL/h (catheter in, because of prolonged labour with haematuria). **Plan**: regular paracetamol and diclofenac, mobilise today, enoxaparin 40 mg SC daily, **keep the catheter for 7 days** in view of the obstructed labour with haematuria, Hb check tomorrow, breastfeeding support, and counselling on the reason for CS and the next birth plan."

The post-CS review checklist ("head to toe, then the 6 Ps"): **P**ulse/BP/temp, **P**ain, **P**allor, **P**ee (urine output, catheter), **P**ad (lochia), **P**ins (calves), plus the fundus, bowel sounds and distension, the wound, breasts and baby, mobilisation, and the drug chart.

10. OSCE Checklist

Station: pre-operative preparation and consent for an emergency CS

Step	Done well when	Critical?
Introduction, identity, rapport	Introduces self, confirms her name and gestation, uses her language or an interpreter	No
States indication and urgency	Gives the indication in one line and names the category with the decision time ("Category 2: within 75 minutes")	Yes
Checks the baby is alive	Listens to the FHR for a full minute after a contraction; reconsiders CS if IUFD	Yes
Explains the procedure	Plain language: spinal, cut low on the abdomen, catheter, baby delivered, stitches	No

Step	Done well when	Critical?
Explains risks	Bleeding/transfusion, infection, bladder/bowel injury, VTE, rare hysterectomy, cut to the baby (1–2%), future rupture and praevia/accreta, gives numbers	Yes
Explains alternatives	E.g. augmentation, ventouse, waiting; what happens if she declines	No
Separate consent for BTL/PPIUD	Asked and consented separately, not assumed	Yes
Asks safety questions	Last meal, allergies, drugs (LMWH, $MgSO_4$), transfusion views, HIV status	Yes
Pre-op orders	Two 16–18 G cannulae, Hb and cross-match (2 or 4 units), Foley, antacid prophylaxis	Yes
Antibiotic timing	Single dose within 60 minutes **before incision**, not after cord clamping	Yes
Theatre checks	Left lateral tilt, repeat FHR, repeat VE if fully dilated, WHO checklist Time out, neonatal resuscitator present	Yes
Next-birth counselling	Mentions spacing (≥18–24 months), incision type recorded, hospital delivery next time	No
Closes	Checks understanding, invites questions, signs consent, thanks her	No

Station: day-1 post-CS review

Step	Done well when	Critical?
Introduction, consent, hand hygiene	As above	No
Asks the daily questions	Pain, eating/drinking, flatus, urine, mobilising, breastfeeding, headache, calf pain, breathlessness, fever, bleeding	No
Vital signs	Pulse, BP, temperature, RR; recognises pulse >100 or fever ≥38.0 °C	Yes
Examines in order	Pallor, chest, breasts, fundus (height and tone), abdomen, wound, lochia, urine, calves (the 6 Ps)	Yes
Recognises PPH	Rising soft fundus or soaked pad → rubs up uterus, calls senior	Yes
Plans ERAS care	Regular multimodal analgesia, mobilise, eat, catheter out when appropriate (or keeps 7–14 days after bladder repair/obstruction), thromboprophylaxis	Yes
Presents	Uses the model post-op presentation structure with a plan	No

ll. Key Points

High-yield points

- Two questions for every CS: **is it indicated? how urgent is it?** Say the category out loud.
- **Category 1 within 30 min; Category 2 within 75 min.**
- **Robson** audits CS rates (Group 5 usually biggest); it does not decide for one woman.
- **Antibiotic before incision**, single dose; prolonged "prophylaxis" breeds resistance.
- **Check the FHR in theatre** and re-examine a fully dilated woman.
- **Lower-segment transverse** is standard; a **classical scar** means elective CS at 36–37 weeks in every future pregnancy.
- **Spinal is the anaesthetic of choice**; not with platelets <75–80 or coagulopathy or uncorrected shock.
- **Abnormal FHR is the earliest sign of rupture** in a woman with a scar.

- **ERAS**: eat, mobilise, catheter out early (but 7–14 days after bladder repair or obstruction with haematuria), thromboprophylaxis.
- **Every woman leaves with a plan for her next birth.**

Numbers & doses

Item	Value
Category 1 decision-to-delivery target	**Within 30 min**
Category 2 decision-to-delivery target	**Within 75 min**
Antibiotic prophylaxis timing	**Single dose within 60 min (15–60 min) before skin incision**
Prophylaxis drug and dose	**Cefazolin 2 g IV** (3 g if ≥120 kg) or **ampicillin 2 g IV**; Uganda: usually **ceftriaxone 1–2 g IV ± metronidazole 500 mg IV**
Repeat prophylactic dose	Surgery longer than about 3–4 h, or blood loss **>1,500 mL**
Cross-match	**2 units** routine emergency CS; **4+ units** for praevia, abruption, rupture, accreta
Platelets for a spinal	Above about **75–80 × 10^9/L**
Bedside clotting test	**2 mL** in a glass tube; no clot by **7 min** = coagulopathy
Spinal drug	**Hyperbaric bupivacaine 0.5%, 2–2.5 mL (10–12.5 mg)** ± fentanyl
Spinal pre-load / co-load	**500–1000 mL** crystalloid
Spinal after prophylactic LMWH	At least **12 h**
Left lateral tilt	**15°**
Aspiration prophylaxis	**Metoclopramide 10 mg IV + omeprazole 40 mg IV**; sodium citrate 0.3 M 30 mL PO
Oxytocin at delivery	**10 IU slow IV or IM** (UK 5 IU); infusion **20 IU in 1 L** at about 40 drops/min for high-risk women
Tranexamic acid for PPH	**1 g IV over 10 min** within 3 h of birth; repeat 1 g if bleeding continues after 30 min
Enoxaparin prophylaxis	**40 mg SC daily** (50–90 kg), first dose **≥4 h after the spinal**; at least **10 days** after emergency CS; 6 weeks if very high risk; UFH alternative **5000 IU SC 12-hourly**
Delayed cord clamping	At least **1 minute** in a vigorous baby
Catheter removal	Mobile and **≥12 h after the last spinal dose**; **7–14 days** after bladder repair or obstructed labour with haematuria
Close subcutaneous fat	Only if **>2 cm** thick
Skin suture removal	**Day 5** (Pfannenstiel); **day 7–10** (midline)
Mobilise	Within **6–12 h**
Uterine rupture risk	**~0.5%** after one LSCS in labour; **4–9%** after a classical scar
VBAC success	**72–75%** (**85–90%** with a previous vaginal birth)
Timing of elective CS	**≥39 weeks**; after a classical scar **36–37 weeks**
Interpregnancy interval	At least **18–24 months** (WHO ≥24 months)
Post-op observations	Every **15 min for the first hour**, then every 30 min for 2 h, then 4-hourly

Item	Value
Urine output	**≥30 mL/h**
SSI surveillance window	**30 days** after surgery
Accreta risk with praevia	About **11%** after 1 CS, **40%** after 2, **>60%** after 3
Uganda CS rate	About **7%** (UDHS 2022); referral hospitals **30–40%**; WHO 10–15%
Fetal laceration at CS	**1–2%**

Exam traps

- **"Give the antibiotic after cord clamping to protect the baby."** Wrong. Give it **before skin incision**; it reduces endometritis and wound infection without harming the baby.
- **"A Pfannenstiel scar means she had a lower-segment CS."** Not necessarily. The skin scar does not tell you the uterine incision: **find the operation note**.
- **"A new abnormal FHR in a woman with a scar is fetal distress."** Think **uterine rupture** first; it is the earliest sign.
- **"Five days of ceftriaxone prevents wound infection."** It does not. Prophylaxis is **one dose**; prolonged courses breed resistance. Treat established infection with a proper course.
- **"Nil by mouth until bowel sounds return."** Outdated. After an uncomplicated CS she can **drink within hours and eat when hungry**.
- **"The category depends on how soon theatre is free."** No. The category describes the **clinical state** of the mother and baby.
- **"Do a CS for an IUFD with a cephalic presentation."** Usually wrong. A **vaginal birth** is safer for her unless there is praevia, transverse lie with obstruction, rupture or another absolute indication.
- **"Induce a woman with a scar with misoprostol."** Contraindicated at term. Use a **Foley catheter**, then ARM and cautious oxytocin at a referral hospital.

Questions seniors ask

Viva questions

1. **Define Category 1 CS and give its decision-to-delivery interval.** An immediate threat to the life of the woman or fetus; deliver as soon as possible, within 30 minutes.
2. **Why give antibiotics before incision rather than after cord clamping?** Tissue levels must be present when contamination occurs. Pre-incision dosing reduces endometritis and wound infection without harming the baby.
3. **What are the Robson groups used for?** A standard, prospective audit of CS rates by 10 mutually exclusive groups, to find which groups drive the rate and to compare facilities.
4. **List 4 indications for a classical CS.** Very preterm with no lower segment; transverse lie back-down with ruptured membranes; large lower-segment fibroid or dense adhesions; anterior praevia/accreta; cervical cancer; perimortem CS.
5. **What are the chances of VBAC success and rupture?** Success is 72–75% (85–90% with a prior vaginal birth). Rupture is about 0.5%, higher with induction or augmentation.
6. **The earliest sign of scar rupture?** An abnormal fetal heart rate (bradycardia or decelerations).
7. **Why is the bladder at risk in CS for obstructed labour?** The bladder is drawn up with the stretched lower segment and is oedematous, so open the peritoneum high and push the bladder down carefully.
8. **When do you remove the catheter after CS?** Once mobile and ≥12 h after the spinal. Keep it 7–14 days after bladder repair or prolonged obstructed labour with haematuria.

9. **Why not close the subcutaneous fat routinely?** It is only beneficial if it is more than 2 cm thick, where it reduces wound separation.
10. **What is the commonest complication of spinal anaesthesia at CS, and how do you manage it?** Hypotension. Left tilt, IV fluids, phenylephrine or ephedrine.

On the ward round

1. **What must you check in theatre just before the incision?** The **fetal heart** (the baby may have died or recovered), and a **repeat VE if she was fully dilated** (the head may now be low enough for a ventouse). Confirm the antibiotic has been given at the Time out.
2. **What are the contraindications to a spinal?** Refusal, **coagulopathy** or platelets below about 75–80, **uncorrected hypovolaemia**, infection at the site, raised intracranial pressure, and some fixed-output cardiac lesions.
3. **It is day 4 after an emergency CS and she has a fever of 38.6 °C. How do you work it up?** Go through the "W" list (womb, wound, water, wind, walking, weaning/breasts, wonder drugs/drip) plus **malaria**. Examine the uterus, lochia, wound, chest, calves, breasts and cannula sites. Send FBC, **blood and pus cultures before antibiotics**, urine dipstick/culture and an RDT. Start therapeutic antibiotics and **drain any pus**.
4. **What category is obstructed labour with a normal FHR? Category 2** (compromise, not immediately life-threatening): deliver within 75 min. With fetal bradycardia or signs of rupture it becomes Category 1.
5. **When would you keep the catheter in longer than usual?** After a **bladder repair** or **prolonged obstructed labour with haematuria**: **7–14 days**, to prevent a vesicovaginal fistula.
6. **What should be written in the operation note?** Indication, category, times, anaesthesia, **incision type (skin and uterus)**, findings, complications, **number of layers of uterine closure**, EBL, counts, and the **plan for the next pregnancy**.
7. **What bloods do you send before an emergency CS? Hb and group & cross-match** for all; HIV test if unknown; add platelets, creatinine, LFTs and clotting (or a bedside clotting test) for PE, abruption, sepsis or IUFD; RDT if febrile.

Mnemonics

- **Urgency 1-2-3-4 = "Life – Compromise – Early – Elective"**; the times to remember are **30 and 75 minutes**.
- **Robson "1-2 first-timers, 3-4 multips, 5 scar, 6-7 breech, 8 twins, 9 lie, 10 preterm."**
- **Pre-op "CABIN-FT"**: **C**onsent, **A**ntibiotics before incision, **B**lood (Hb and cross-match), **I**V lines ×2, **N**il by mouth/antacid, **F**oley catheter, **T**ilt (and **T**heatre FHR check).
- **Layers to open ("Some Fat Sheaths Rarely Protect Uteri")**: Skin, Fat, Sheath, Recti (separate), Peritoneum, Uterus (after the uterovesical fold and bladder).
- **Signs of rupture ("FHR FIRST")**: **F**etal heart abnormal, **H**ypotension/tachycardia, **R**eceding presenting part, **F**etal parts palpable, **I**ncreasing constant pain, **R**ed urine (haematuria), **S**top in contractions, **T**ip of shoulder pain.
- **TOLAC candidate "ONE-LOW-HEAD-HOSPITAL"**: **one** previous CS, **low** transverse incision, cephalic **head**, delivering in a **hospital** (or HC IV) with a theatre.
- **Post-CS fever "W" list**: Womb, Wound, Water, Wind, Walking, Weaning (breasts), Wonder drugs/drip, plus malaria.
- **Post-CS review "6 Ps"**: Pulse, Pain, Pallor, Pee, Pad, Pins.

OSCE tips

> **EXAM FAVOURITE**
> OSCE stations on CS usually test **consent** (explain indication, procedure, risks and alternatives; check understanding; allow questions), **post-op review**, or **counselling for VBAC**. Always start with an introduction, confirm identity, **explain the reason** in plain language, avoid jargon, give numbers ("about 1 in 200"), and close by checking understanding and next steps.

- **VBAC counselling**: candidate criteria, 72–75% success (85–90% with a prior vaginal birth), rupture about 1 in 200 (2–3 times higher with induction or augmentation), about 1 in 4 need an emergency CS, deliver at a facility with theatre and blood, no misoprostol.
- Use the OSCE checklists in section 10.

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SELF-TEST

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define the four urgency categories of caesarean section, with target decision-to-delivery intervals.	**Cat 1**: immediate threat to the life of the woman or fetus. Deliver ASAP, **within 30 min**. **Cat 2**: maternal or fetal compromise, not immediately life-threatening. **Within 75 min**. **Cat 3**: no compromise but needs early delivery. **Cat 4**: elective, at a time to suit the woman and the team.
2	Which six variables define the Robson Ten-Group Classification?	**Parity**, **previous CS**, **onset of labour** (spontaneous, induced, CS before labour), **gestational age** (<37 or ≥37 wk), **number of fetuses**, and **presentation/lie**.
3	Robson Group 5 contains which women?	All **multiparous women with at least one previous CS**, single cephalic, ≥37 weeks. It is usually the largest contributor to a hospital's CS rate.
4	When should prophylactic antibiotics be given for CS, and why?	A **single IV dose within 60 minutes before skin incision** (not after cord clamping). Tissue levels are then present when contamination occurs, which reduces endometritis and wound infection without harming the baby.
5	WHO/ACOG first-choice antibiotic prophylaxis for CS, and the usual Ugandan practice.	**Cefazolin 2 g IV** (3 g if ≥120 kg) or **ampicillin 2 g IV**. In Uganda most hospitals give **ceftriaxone 1–2 g IV ± metronidazole 500 mg IV** at induction of anaesthesia. It is a single dose, not a 5-day course.
6	Why is the lower segment preferred for the uterine incision?	It is **thin, less vascular and hardly contracts**, so there is less bleeding and better healing. It is covered by loose peritoneum, so the bladder can be pushed down. The later rupture risk is about **0.5%**, versus **4–9%** for a classical scar.
7	List 5 indications for a classical (upper-segment) CS.	1. Very preterm with no formed lower segment 2. Transverse lie, back down, with ruptured membranes or arm prolapse 3. Large lower-segment fibroid or dense adhesions 4. Anterior placenta praevia with large vessels / accreta 5. Invasive cervical cancer 6. Perimortem CS
8	What are the implications of a previous classical CS for the next pregnancy?	**No labour.** Rupture risk is 4–9%, and rupture can happen before labour. Plan an **elective CS at about 36–37 weeks**, delivered at a hospital. Document this clearly on her card.
9	Chance of a successful VBAC, and the risk of uterine rupture during TOLAC.	Success is **72–75%** (85–90% if she has had a previous vaginal birth). Rupture is **about 0.5% (1 in 200)**, rising 2–3 fold with induction or augmentation (RCOG GTG 45).
10	Absolute contraindications to a trial of labour after CS.	Previous **classical, T or J incision**; previous **uterine rupture**; **placenta praevia** or any other indication for CS; a **facility without CS/transfusion capability**. Two or more previous CS is a contraindication in most Ugandan units.
11	What is the earliest and most consistent sign of uterine scar rupture?	An **abnormal fetal heart rate** (bradycardia or prolonged/recurrent decelerations). Others: maternal tachycardia/shock, constant pain, shoulder-tip pain, cessation of contractions, loss of station, haematuria, vaginal bleeding.

#	Question	Answer
12	Which induction agent is contraindicated in a woman with a uterine scar at term?	**Misoprostol** (and prostaglandins in general, in most guidance), because of the high rupture risk. Use **Foley catheter ripening, then ARM, then cautious oxytocin** at a referral hospital, or offer ERCS.
13	Difference between scar dehiscence and uterine rupture.	**Dehiscence**: the scar separates but the **serosa is intact**. It is usually silent, a window found at repeat CS. **Rupture**: a **full-thickness** tear into the peritoneal cavity, with haemorrhage and fetal compromise. It is an emergency.
14	Joel-Cohen vs Pfannenstiel incision.	**Joel-Cohen**: straight transverse incision about 3 cm above the symphysis, then mostly **blunt** dissection. It is faster, with less blood loss, pain and fever (NICE preferred). **Pfannenstiel**: curved incision 2 fingerbreadths above the symphysis, with sharp dissection. Good cosmesis.
15	Why tilt the woman 15° to the left for CS?	To relieve **aortocaval compression** by the gravid uterus. Otherwise venous return falls, and with the sympathetic block of a spinal this causes severe hypotension and fetal hypoxia.
16	Spinal anaesthesia for CS: usual drug and dose, and the commonest complication.	**Hyperbaric bupivacaine 0.5%, 2–2.5 mL (10–12.5 mg)** ± fentanyl, at L3/4 or L4/5. The commonest complication is **hypotension**: treat with left tilt, IV fluid and **phenylephrine 50–100 µg** or **ephedrine 6 mg** boluses.
17	When should the urinary catheter be removed after CS?	Once she is **mobile and ≥12 h after the last spinal dose**. Keep it **7–14 days** after bladder repair or after prolonged obstructed labour with haematuria (to prevent a fistula).
18	Five ERAS elements after CS.	1. **Early oral intake** (drink within hours, eat when hungry) 2. **Early mobilisation** (6–12 h) 3. **Early catheter removal** 4. **Multimodal regular analgesia** (paracetamol + NSAID ± opioid) 5. **VTE prophylaxis** and no unnecessary post-op antibiotics
19	Thromboprophylaxis after emergency CS: drug, dose, duration.	**Enoxaparin 40 mg SC once daily** (weight 50–90 kg; 20 mg if <50 kg; 60 mg if 91–130 kg), started 4–6 h post-op. Give for **at least 10 days** (6 weeks if high risk). If LMWH is unavailable: **UFH 5000 IU SC 12-hourly**. Plus early mobilisation and hydration.
20	What must every woman know before discharge after CS?	Why she had the CS; the **type of uterine incision**; the **plan for her next birth** (HC IV/hospital; TOLAC vs ERCS); **spacing ≥18–24 months** and contraception (PPIUD, implant, BTL); danger signs; and review at 6 days and 6 weeks.
21	List the pre-op sign-off items before a woman leaves the ward for an emergency CS.	1. **Indication and urgency category** written, with the decision time 2. **Consent** (plus separate BTL/PPIUD consent) 3. **Anaesthetic review** (airway, fasting, platelets, clotting, anticoagulants) 4. **Hb and group & cross-match** (2 units; 4+ for praevia, abruption, rupture, accreta) 5. Two **16–18 G cannulae** 6. **Foley catheter** 7. **Antacid prophylaxis** 8. **Antibiotic within 60 min before incision** 9. **HIV status** and eMTCT plan 10. **Neonatal resuscitator** called 11. **WHO Surgical Safety Checklist**, with a repeat FHR check (and VE if fully dilated) in theatre

#	Question	Answer
22	How do you perform and interpret the bedside clotting test?	Put **2 mL of venous blood** in a small, clean, dry **plain glass tube**, hold it in your closed fist to keep it warm, tip it after **4 min**, then every minute. **No clot by 7 min**, or a soft clot that breaks down, means **coagulopathy** (e.g. DIC in abruption). Consequence: **no spinal**; prepare fresh whole blood/FFP and expect PPH.
23	Day-1 post-CS ward review: what do you check?	**Vital signs** (pulse, BP, temp, RR), then the **6 Ps**: **P**ulse/BP/temp, **P**ain, **P**allor, **P**ee (urine ≥30 mL/h, colour, catheter), **P**ad (lochia amount and smell), **P**ins (calves). Plus the **fundus** (firm, at or below the umbilicus), abdomen (distension, bowel sounds), **wound** (dressing off at 24–48 h), chest, breasts and baby, mobilisation, eating, and the **drug chart** (analgesia, thromboprophylaxis, no unnecessary antibiotics).
24	Causes of fever after CS, by typical day (the "W" list).	**Wind** (chest/atelectasis): day 1–2 **Womb** (endometritis): day 2–5 **Water** (UTI, catheter): day 3–5 **Wound** (SSI): **day 4–7 Walking** (DVT/PE): day 5 onwards **Weaning** (breast engorgement/mastitis): day 3 onwards **Wonder drugs / drip** (drug fever, cannula phlebitis): any time In Uganda, always add **malaria** (RDT or smear).
25	Define surgical site infection (SSI) and its three levels. How do you manage a superficial SSI after CS?	Infection **within 30 days** of surgery. **Superficial incisional** (skin and fat), **deep incisional** (fascia and muscle), **organ/space** (endometritis, pelvic abscess). Superficial SSI: **pus swab for culture before antibiotics**, **open the wound at the point of discharge** (remove 1–2 sutures) to drain, daily dressing, antibiotics only if there is **cellulitis or systemic illness**, guided by culture (Ugandan isolates are often ceftriaxone-resistant). Report clusters to IPC.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 29-year-old P1+0 with one previous lower-segment CS is on a trial of scar at a regional referral hospital. She is 6 cm dilated. The midwife calls you: the fetal heart has been 80 bpm for 4 minutes, and the woman complains of constant lower abdominal pain. Her pulse is 118 bpm and her BP is 100/60 mmHg. What is the MOST appropriate action?

A Classify as Category 3 and review her progress in 2 hours
B Turn her to the left lateral, give oxygen, and continue the trial of scar with auscultation every 15 minutes
C Declare a Category 1 laparotomy/CS for suspected uterine rupture and aim to deliver within 30 minutes
D Classify as Category 2 and request an ultrasound scan to confirm rupture before theatre
E Apply a vacuum extractor to expedite delivery

2. A 25-year-old primigravida is going to theatre for an emergency CS for fetal distress at Mengo Hospital. The anaesthetic officer asks when to give the ceftriaxone. What is the BEST answer?

A A single IV dose within 60 minutes before the skin incision
B After the cord is clamped, to avoid exposing the baby to the antibiotic
C A 5-day IV course starting after surgery
D Oral amoxicillin for 7 days after discharge
E Only if the membranes have been ruptured for more than 18 hours

3. A 32-year-old P2+0 (both vaginal births) has a singleton cephalic pregnancy at 38 weeks. She is induced for pre-eclampsia and eventually delivered by CS for fetal distress. In the monthly Robson audit, which group does she belong to?

A Group 1
B Group 2a
C Group 3
D Group 4a
E Group 5

4. A 30-year-old G2P1 books at 16 weeks. Her discharge summary from 2 years ago shows a classical CS at 28 weeks for a transverse lie with ruptured membranes. She wants a normal delivery this time. What is the BEST plan?

A Trial of labour at a regional referral hospital with continuous monitoring
B Elective repeat CS at about 36–37 weeks, before the onset of labour
C Induction with misoprostol 25 µg at 39 weeks to avoid a long labour
D Await spontaneous labour at her nearest HC IV
E Elective CS at 41 weeks if she has not laboured

5. Four hours after an emergency CS for obstructed labour, a 22-year-old is restless. Her pulse is 124 bpm, BP 86/50 mmHg, and she is pale. The abdomen is increasingly distended and tender. The uterus is firm at the umbilicus and the lochia is scanty. The oxytocin infusion is running. What is the MOST appropriate next step? Advanced

A Increase the oxytocin infusion rate and massage the uterus
B Give misoprostol 800 µg sublingually
C Observe and repeat the haemoglobin tomorrow morning
D Start IV ceftriaxone and metronidazole for sepsis
E Resuscitate with IV fluids and blood, and return her to theatre for re-laparotomy

6. A 28-year-old had an uncomplicated elective CS under spinal anaesthesia 3 hours ago. She is comfortable, has no nausea, and asks for a cup of tea. What is the MOST appropriate response?

A Keep her nil by mouth until bowel sounds are heard
B Keep her nil by mouth for 24 hours and continue IV fluids
C Allow oral fluids now and a light diet as tolerated
D Allow only IV fluids until she passes flatus
E Insert a nasogastric tube to prevent ileus

7. A 19-year-old primigravida had a CS for obstructed labour after 2 days in labour at home. She had haematuria on catheterisation. At surgery the bladder was oedematous and high but intact. When should the urinary catheter be removed? Advanced

A At 12 hours, once the spinal has worn off
B After continuous drainage for 7–14 days
C In theatre at the end of the operation
D At 24 hours, if the urine is clear
E Clamp it intermittently and remove it when she feels the urge to void

8. A 31-year-old P1+0 with one previous lower-segment CS (for breech) is 41+3 weeks at a regional referral hospital. The cervix is closed, long and firm (Bishop score 3). The fetus is cephalic with normal growth. She strongly wishes to try for a vaginal birth. Which is the BEST option? Advanced

A Misoprostol 25 µg vaginally every 6 hours until labour starts
B High-dose oxytocin infusion from the start to shorten the induction
C Wait until 43 weeks for spontaneous labour
D Counsel her; offer ERCS or mechanical ripening with a Foley catheter, then ARM and cautious oxytocin under senior supervision
E Discharge her to deliver at her local HC III, which is closer to home

9. Five minutes after a spinal anaesthetic for elective CS, a 27-year-old becomes nauseated and pale. Her BP is 78/40 mmHg (baseline 120/75) and the FHR is 100 bpm. She is lying flat. What is the MOST appropriate immediate management?

A Left uterine displacement/tilt, a rapid IV crystalloid bolus and an IV vasopressor (phenylephrine or ephedrine)
B Convert immediately to general anaesthesia
C Give adrenaline 1 mg IV
D Give furosemide 40 mg IV
E Put her in a steep head-down position and wait for the BP to recover

10. On day 3 after an emergency CS for prolonged labour with ruptured membranes for 30 hours, a 24-year-old has a temperature of 38.9 °C and pulse of 112 bpm. Her lochia is foul-smelling and the uterus is tender and subinvoluted. The wound is clean and the malaria RDT is negative. What is the MOST appropriate management?

A Oral amoxicillin 500 mg 8-hourly and review in a week

B Paracetamol and observation, as post-op fever on day 3 is expected

C Immediate re-laparotomy and hysterectomy

D Artemether-lumefantrine because malaria is common

E IV ampicillin + gentamicin + metronidazole (or ceftriaxone + metronidazole), take cultures, and check for retained products

11. A 26-year-old at 27 weeks has a transverse lie, back down, with ruptured membranes and a prolapsed fetal arm. The fetus is alive. The lower segment is poorly formed on ultrasound. Which uterine incision is MOST appropriate? Advanced

A Low transverse lower-segment incision through a Pfannenstiel

B Internal podalic version followed by breech extraction

C Vertical upper-segment (classical) incision

D Destructive operation (embryotomy) vaginally

E Push the arm back and await spontaneous version

12. A 34-year-old at 36 weeks has severe pre-eclampsia with HELLP syndrome (platelets 45 × 10^9/L) and a pathological fetal heart. Her BP is controlled on hydralazine and she has had magnesium sulphate. A Category 2 CS is decided. Which anaesthetic technique is MOST appropriate? Advanced

A Spinal anaesthesia with hyperbaric bupivacaine

B Epidural anaesthesia

C Local infiltration anaesthesia alone

D General anaesthesia with rapid sequence induction and blunting of the pressor response to intubation

E Delay surgery for 24 hours to transfuse platelets until the count exceeds 100

13. A 28-year-old P2+0 had a vaginal birth 5 years ago and then a lower-segment CS 2 years ago for breech presentation. She is now 37 weeks with a cephalic singleton and asks what her chance of a vaginal birth is if she labours. What is the BEST estimate?

A About 25%

B About 85–90%

C About 50%

D About 72–75%

E Over 99%

14. During a CS at full dilatation after a 3-hour second stage, the fetal head is deeply impacted in the pelvis and you cannot lift it into the incision. What is the BEST next step? Advanced

A Ask an assistant to push the head up from the vagina while you flex the head and lift it, and call for senior help

B Apply strong fundal pressure while pulling on the baby's shoulders

C Immediately extend the incision into an inverted-T before attempting disimpaction

D Pull firmly on the baby's jaw and neck to lever the head out

E Close the uterus and attempt vaginal ventouse delivery

15. A 36-year-old P3, weight 82 kg (BMI 34), had an emergency CS in labour complicated by a 1.2 L PPH. Bleeding has now settled. What is the MOST appropriate thromboprophylaxis?

A No pharmacological prophylaxis; early mobilisation is enough
B Aspirin 75 mg daily for 6 weeks
C Warfarin starting on day 1
D Therapeutic-dose enoxaparin 1 mg/kg twice daily
E Enoxaparin 40 mg SC once daily from 4–6 hours post-op for at least 10 days, plus early mobilisation

Answers & explanations

1. C · Declare a Category 1 laparotomy/CS for suspected uterine rupture and aim to deliver within 30 minutes

Sustained fetal bradycardia plus constant pain and maternal tachycardia in a woman with a scar means **uterine rupture until proven otherwise**. This is an immediate threat to life, so it is **Category 1** (within 30 min). Resuscitate while moving to theatre. **A** is wrong: there is compromise, so Category 3 is inappropriate. **B**: intrauterine resuscitation is reasonable for a few seconds, but continuing the trial with a bradycardia and these signs is dangerous. **D**: rupture is a clinical diagnosis, and waiting for a scan wastes minutes the baby does not have. **E**: vacuum requires full dilatation; she is 6 cm.

2. A · A single IV dose within 60 minutes before the skin incision

Evidence (Cochrane, WHO, NICE, and a Mulago RCT) shows that **pre-incision** prophylaxis reduces endometritis and wound infection compared with dosing after cord clamping, with no neonatal harm. So **A** is correct. **B** is the outdated practice. **C** and **D**: prolonged post-op courses do not prevent SSI and drive resistance (the WHO SSI guideline recommends against them). **E**: every CS needs prophylaxis, not only those with prolonged ROM.

3. D · Group 4a

She is **multiparous without a previous CS**, single cephalic, ≥37 weeks, with **induced** labour. That is **Group 4a**. **Groups 1 and 2** are nulliparous women. **Group 3** is spontaneous labour. **Group 5** requires a previous CS. The final mode of delivery does not change the group: groups are defined by characteristics at admission.

4. B · Elective repeat CS at about 36–37 weeks, before the onset of labour

A **classical scar** has a 4–9% rupture risk and can rupture **before labour**. TOLAC is contraindicated, so plan an **elective CS at 36–37 weeks** (**B**). **A** and **D** allow labour, which is contraindicated. **C**: misoprostol with any uterine scar at term is contraindicated, and doubly so with a classical scar. **E**: waiting to 41 weeks invites spontaneous labour and rupture.

5. E · Resuscitate with IV fluids and blood, and return her to theatre for re-laparotomy

Shock with a **firm uterus and little vaginal bleeding** plus a distending abdomen means **concealed intra-abdominal (or broad-ligament) haemorrhage**, often from a uterine angle extension or a slipped ligature. The treatment is **resuscitation and re-laparotomy** (**E**). **A** and **B** treat atony, but the uterus is contracted. **C** is dangerous delay. **D**: sepsis 4 h post-op is unlikely, and antibiotics do not stop bleeding.

6. C · Allow oral fluids now and a light diet as tolerated

ERAS and NICE: women recovering well after CS can **eat and drink when they feel hungry or thirsty**. Early feeding shortens recovery and does not increase ileus. **A, B and D** are outdated rituals that prolong IV fluids, hunger and admission. **E**: an NG tube is only for established obstruction or ileus.

7. B · After continuous drainage for 7–14 days

Prolonged obstructed labour compresses the bladder base between the head and the pubis, which causes **ischaemic pressure necrosis**. A **vesicovaginal fistula** may appear 5–10 days later. **Continuous free drainage for 7–14 days** (**B**) rests the bladder and can prevent or heal a small fistula. **A, C and D** are appropriate after routine CS but not after obstruction with haematuria. **E**: clamping lets the bladder distend, which harms the ischaemic bladder.

8. D · Counsel her; offer ERCS or mechanical ripening with a Foley catheter, then ARM and cautious oxytocin under senior supervision

Induction with a scar increases the rupture risk 2–3 fold, and **prostaglandins such as misoprostol are contraindicated** (**A**). The safest options are **ERCS or mechanical (Foley) ripening, then ARM and carefully titrated oxytocin** in a hospital with a theatre, after counselling (**D**). **B**: high-dose oxytocin raises the rupture risk. **C**: prolonged post-term pregnancy increases stillbirth. **E**: an HC III cannot do CS, so a woman with a scar must deliver at an HC IV or hospital.

9. A · Left uterine displacement/tilt, a rapid IV crystalloid bolus and an IV vasopressor (phenylephrine or ephedrine)

Spinal-induced **hypotension** (sympathetic block plus aortocaval compression) is common and reduces uteroplacental flow, hence the fetal bradycardia. Treat it with **left tilt, fluids and a vasopressor** (**A**). **B**: GA adds airway risk and doesn't treat the cause. **C**: adrenaline 1 mg is a cardiac-arrest dose. **D**: a diuretic worsens hypovolaemia. **E**: steep head-down can spread a hyperbaric spinal higher (a high block) and does not relieve caval compression.

10. E · IV ampicillin + gentamicin + metronidazole (or ceftriaxone + metronidazole), take cultures, and check for retained products

Fever, foul lochia and a tender subinvoluted uterus after a CS in prolonged labour mean **post-caesarean endometritis (puerperal sepsis)**. Take cultures, then give **IV broad-spectrum antibiotics** until she has been afebrile for 48 hours (**E**). Look for retained products and a pelvic abscess. **A** is inadequate for sepsis. **B**: fever after 48 h is not physiological. **C**: surgery is reserved for abscess, necrotising infection or failure of medical therapy. **D**: the RDT is negative and the uterine signs explain the fever.

11. C · Vertical upper-segment (classical) incision

A back-down transverse lie with ruptured membranes, very preterm, with an **unformed lower segment** is a classic indication for a **classical (vertical upper-segment)** incision (**C**): the fetus cannot be safely extracted through a small transverse incision. **A** risks extension into the uterine vessels and a trapped fetus. **B**: internal podalic version is reserved for a second twin, and it causes rupture in a singleton with ruptured membranes. **D**: the fetus is alive. **E**: a transverse lie with a prolapsed arm will not correct spontaneously and obstructs.

12. D · General anaesthesia with rapid sequence induction and blunting of the pressor response to intubation

A platelet count below about 75–80 × 10^9/L is a contraindication to **neuraxial** anaesthesia (**A, B**) because of the risk of spinal/epidural haematoma. **GA with rapid sequence induction** (**D**), with measures to blunt the hypertensive surge at intubation (for example IV labetalol or an opioid), is appropriate. **C**: local infiltration is a last resort when no anaesthetist is available. **E**: the fetus is compromised, and HELLP worsens with time. Delay is dangerous.

13. B · About 85–90%

Overall VBAC success is 72–75% (**D**), but **a previous vaginal birth** raises it to about **85–90%** (**B**). A non-recurrent indication (breech) is also favourable. **A** and **C** underestimate the chance, which could scare her into an unnecessary ERCS. **E** overstates it: about 1 in 10 still need an emergency CS.

14. A · Ask an assistant to push the head up from the vagina while you flex the head and lift it, and call for senior help

The **push method** (an assistant disimpacts the head vaginally with a cupped hand) or the **pull method** (reverse breech extraction) are the recognised techniques, and a senior should be called early (**A**). **B**: the shoulders are not accessible, and fundal pressure pushes the head further down. **C**: an inverted-T commits her to CS in all future pregnancies and is only for failed disimpaction. **D**: levering on the jaw or neck causes fetal injury and uterine extensions. **E** is irrational: the head could not be delivered vaginally, which is why she is in theatre.

15. E · Enoxaparin 40 mg SC once daily from 4–6 hours post-op for at least 10 days, plus early mobilisation

She has several VTE risk factors: CS in labour, age >35, obesity and PPH >1 L. By RCOG GTG 37a she needs **prophylactic LMWH (enoxaparin 40 mg for 50–90 kg) for at least 10 days**, started once haemostasis is secure (**E**). **A** under-treats a high-risk woman. **B**: aspirin is not recommended for postpartum VTE prophylaxis. **C**: warfarin is for treatment, not routine prophylaxis. **D** is a treatment dose and would increase bleeding.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Obstructed labour at an HC IV

A 19-year-old primigravida at term is brought to an HC IV in Mukono district after 2 days of labour. She pushed at home with a traditional birth attendant for 'many hours'. She is exhausted and dehydrated. Pulse 116 bpm, BP 110/70 mmHg, temp 37.8 °C. She has a tense lower abdomen with a groove running obliquely below the umbilicus. The head is 3/5 palpable, FHR 168 bpm, the vulva is oedematous, and there is dark urine on catheterisation.

1. What further history do you want, and why?

- **Duration of labour and of pushing**, and time of ROM (sepsis risk, obstruction)
- **Herbs or injections** given at home (herbal oxytocics or oxytocin cause hyperstimulation and rupture)
- **Fetal movements** (still alive?)
- **ANC**: HIV status, Hb, blood group, height (short stature suggests a contracted pelvis)
- **Last meal** (aspiration risk)
- **Allergies**
- Any **previous surgery**

2. Interpret the examination findings. What is your diagnosis?

Obstructed labour with **impending uterine rupture**. The oblique groove is a **Bandl's ring** (a pathological retraction ring). The head is 3/5 palpable after prolonged pushing (no descent), there is maternal tachycardia, low-grade fever and dehydration, fetal tachycardia (compromise or infection), an oedematous vulva, and **haematuria** from bladder compression. On VE, expect 3+ moulding, big caput and an oedematous cervix.

3. What are your immediate management steps before theatre?

1. Call the MO/obstetrician and anaesthetic officer: **Category 2 CS** (Category 1 if the FHR becomes bradycardic or signs of rupture appear).
2. **No oxytocin, no fundal pressure, no ventouse** (the head is high).
3. Two **16–18 G cannulae**; **Hb, cross-match 2 units**, HIV test if unknown.
4. **Ringer's lactate 1 L fast**, then titrate to urine output.
5. **Leave the catheter on free drainage.**
6. Antibiotics: she has a fever and prolonged ROM, so give a **therapeutic regimen**, not just prophylaxis: ampicillin 2 g IV + gentamicin 5 mg/kg IV + metronidazole 500 mg IV (or ceftriaxone 2 g + metronidazole 500 mg IV) before incision.
7. Metoclopramide 10 mg IV (aspiration prophylaxis). Consent. Left tilt. HBB-trained person for the baby.

4. What operative difficulties do you anticipate and how will you avoid them?

- **High, oedematous bladder**: open the parietal peritoneum **high**, and identify and push the bladder down gently.
- **Very thin lower segment**: incise carefully to avoid cutting the baby. **Extend bluntly** and watch the angles (lateral extensions into the uterine vessels and ureter).
- **Impacted head**: the assistant disimpacts from the vagina (push method) or use reverse breech extraction (pull). Call a senior early.
- **PPH from atony** after a long labour: oxytocin 10 IU IV, then an infusion of 20 IU in 1 L; TXA if bleeding.
- **Infection**: careful asepsis, peritoneal toilet.

5. At surgery the uterus is intact. A live 3.4 kg male is delivered (Apgar 5 and 8). EBL 800 mL. What is your post-operative plan?

- Obs every 15 min for 1 h, then 4-hourly; uterine tone and lochia; **oxytocin infusion 20 IU in 1 L** over 4–8 h.
- **Catheter on free drainage for 7–14 days** (risk of VVF from pressure necrosis); record fluid balance.
- **Continue IV antibiotics** (therapeutic course) until afebrile for 48 h, then switch to oral.
- Analgesia: paracetamol 1 g 6-hourly + diclofenac 50 mg 8-hourly (if renal function is fine) ± tramadol.
- **Enoxaparin 40 mg SC daily** (or UFH 5000 IU 12-hourly) once haemostasis is secure; early mobilisation.
- Hb on day 1; iron/folate.
- Baby: neonatal review for sepsis risk (prolonged ROM, maternal fever), and start breastfeeding.

6. What will you tell her before discharge about future pregnancies?

She had a CS because the baby's head could not pass (the obstruction may recur if her pelvis is small). She has a **lower-segment scar**. Next time she must **book early and deliver at an HC IV or hospital**, never at home. A trial of scar may be possible depending on the pelvic assessment, but the obstetrician will decide at 36 weeks. **Wait at least 18–24 months before the next pregnancy**, so offer contraception (implant, injectable or LAM). Report urine leakage immediately (a fistula can appear 1–2 weeks later). Give danger signs.

Take-away: Obstructed labour needs resuscitation, a catheter and antibiotics before a CS. Expect a high bladder and a thin lower segment. Keep the catheter for 7–14 days after haematuria. And send every woman home with a birth plan for next time.

Case 2: Trial of scar that goes wrong

A 27-year-old P1+0 with one previous emergency LSCS (3 years ago for 'failure to progress') is admitted in spontaneous labour at 39 weeks to a regional referral hospital. On admission she is 4 cm dilated, the head is 3/5 palpable, and FHR is 140 bpm. Six hours later she is still 5 cm dilated with the head 3/5, contractions are 4 in 10 minutes, and she complains of pain over the scar between contractions.

1. What is your assessment of this trial of scar?

The trial is **failing**. Six hours of adequate contractions (4 in 10) have produced only 1 cm of dilatation with **no descent**. The previous indication was a **recurrent** one (failure to progress, possibly CPD), which predicts TOLAC failure. **Scar pain between contractions** is a warning sign of dehiscence. The partograph has crossed the **alert line** and is heading for the action line.

2. What would you do now, and what would you NOT do?

> **Do**: inform the senior, recheck the FHR for a full minute after a contraction, check maternal pulse and BP, look for vaginal bleeding and haematuria, and prepare for a **Category 2 repeat CS** (consent, IV access, cross-match 2 units, antibiotics before incision). **Do NOT**: start oxytocin augmentation for poor progress in a scarred uterus with suspected CPD; do not continue 'waiting and seeing'.

3. While preparing theatre, the FHR drops to 70 bpm and does not recover. Her pulse rises to 124 bpm and contractions stop. The head is now 4/5 palpable. What has happened and what do you do?

> **Uterine rupture**: fetal bradycardia, maternal tachycardia, cessation of contractions and **loss of station**. **Category 1 laparotomy**: call everyone, stop any oxytocin, give oxygen, use a second large-bore cannula, cross-match **4 units**, give warm crystalloid, catheterise (look for haematuria), and give antibiotics. Go to theatre immediately; a midline incision is reasonable.

4. At laparotomy the scar has ruptured with a lateral extension into the left broad ligament, and the baby is in the abdomen (fresh stillbirth). She is hypotensive despite 2 units of blood. What are the surgical options and how do you choose?

> - **Repair** the uterus in two layers if the tear is clean, she is stabilising, and she wants more children.
> - **Subtotal (or total) hysterectomy** if the tear is extensive (broad ligament, uterine vessels, cervix), if bleeding cannot be controlled, if the tissues are infected or necrotic, or if she remains unstable.
>
> Here, with a broad-ligament extension and ongoing instability, **hysterectomy is life-saving** (life before fertility). Before closing, identify the **ureter** (at risk in broad-ligament haematomas) and check the **bladder**.

5. What post-operative care and follow-up does she need?

> - HDU monitoring, transfusion to Hb above about 8 g/dL, and a therapeutic antibiotic course; thromboprophylaxis once bleeding is controlled.
> - **Bereavement care**: let her see and hold the baby if she wishes, psychosocial support, and lactation suppression advice (e.g. cabergoline 1 mg single dose if available, a supportive bra).
> - Honest **open disclosure** and counselling (she cannot carry another pregnancy after hysterectomy).
> - **MPDSR**: this perinatal death must be reviewed (audit of TOLAC selection and monitoring, and response time).

> **Take-away:** TOLAC needs good selection (a recurrent indication such as CPD lowers success), a partograph, and vigilance. Poor progress in a scarred uterus means CS, not oxytocin. An abnormal FHR with loss of station and cessation of contractions means rupture: Category 1 laparotomy.

Case 3: Fever on day 3 after CS at Mengo

A 30-year-old P3 is on day 3 after an emergency CS for fetal distress (membranes ruptured 20 hours before surgery). She received ceftriaxone 2 g IV at induction. Today her temperature is 38.6 °C and pulse 108 bpm. She complains of lower abdominal pain and feeling unwell. She is breastfeeding.

1. What are the possible sources of this post-operative fever? (Think systematically.)

> Use the **'post-CS fever' checklist: Womb, Wound, Water, Wind, Walking, Wonder drugs, Breasts, and Malaria**.
>
> - **Womb**: endometritis (commonest after CS in labour with prolonged ROM); retained products; pelvic abscess
> - **Wound**: SSI, haematoma, abscess, necrotising fasciitis
> - **Water**: UTI or catheter-associated infection
> - **Wind**: atelectasis or pneumonia (especially after GA)
> - **Walking**: DVT/PE
> - **Wonder drugs**: drug fever, thrombophlebitis at the cannula site
> - **Breasts**: engorgement or mastitis (usually later)
> - **Malaria** (always in Uganda)

2. What focused examination will you perform?

> - Vitals including **RR, SpO_2 and BP** (sepsis screen: RR ≥22, SBP ≤100, altered mentation)
> - Pallor; **breasts**; **chest** (crepitations)
> - **Abdomen**: distension, bowel sounds, fundal height (subinvolution), **uterine tenderness**, peritonism
> - **Wound**: remove the dressing and look for erythema, induration, discharge, gaping; gently probe for pus
> - **Lochia**: smell, colour, amount
> - **Calves**: swelling, tenderness
> - **Cannula sites**; urine (catheter still in?)

3. She has foul lochia, a tender bulky uterus, and a wound with 2 cm of surrounding erythema but no discharge. What investigations do you order?

> - **FBC** (WBC, Hb)
> - **Blood culture** before antibiotics
> - **High vaginal / endocervical swab** for culture and sensitivity
> - **Malaria RDT or blood smear**
> - **Urinalysis and urine culture**
> - **Pelvic ultrasound** (retained products, pelvic collection, wound or rectus sheath haematoma)
> - Renal function and lactate if septic
> - CRP if available

4. Outline your management.

> **Diagnosis: post-caesarean endometritis with early wound cellulitis.**
>
> 1. **IV antibiotics**: ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV once daily + metronidazole 500 mg IV 8-hourly (or ceftriaxone 2 g IV daily + metronidazole 500 mg IV 8-hourly). Adjust to cultures: Mulago data show many **ceftriaxone-resistant *Klebsiella/E. coli* and MRSA**, so escalate (e.g. to meropenem or vancomycin per culture) if she fails to respond in 48–72 h.
> 2. Continue until **afebrile for 48 hours**, then oral (amoxicillin-clavulanate ± metronidazole) to complete about 7 days total for the wound.
> 3. **Paracetamol**, fluids, and continue breastfeeding (these antibiotics are compatible).
> 4. **Evacuate retained products** if found; **drain** any wound or pelvic collection.
> 5. **Thromboprophylaxis** (infection raises the VTE risk).

5. On day 5 of treatment she is still febrile despite antibiotics, and the wound is now draining pus. What do you consider?

> - **Wound abscess or infected haematoma**: open the wound, drain it, send pus for culture, and dress daily. Secondary closure or healing by secondary intention.
> - **Pelvic or subsheath abscess**: ultrasound; drainage (surgical or guided).
> - **Resistant organism**: change antibiotics according to culture.
> - **Septic pelvic thrombophlebitis** (fever despite antibiotics with no collection): a diagnosis of exclusion, treated with anticoagulation.
> - **Necrotising fasciitis** (pain out of proportion, skin discolouration, crepitus): emergency debridement.

Take-away: Post-CS fever needs a systematic search (womb, wound, water, wind, walking, drugs, breasts and malaria). Endometritis is commonest after CS in labour with prolonged ROM. Treat it with IV triple therapy, culture-guided escalation (resistance is common in Uganda), and drainage of any collection.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.